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October 15, 2026

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Abstract

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Chapter 1

Introduction

1.1 Background to the study

A very brief background to your area of research. Start off with a general introduction to the area and then narrow it down to your focus area. Used to set the scene [**?**].

1.2 Objectives of this study

1.2.1 Problems to be investigated

Description of the main questions to be investigated in this study.

1.2.2 Purpose of the study

Give the significance of investigating these problems. It must be obvious why you are doing this study and why it is relevant.

1.3 Scope and Limitations

Scope indicates to the reader what has and has not been included in the study. Limitations tell the reader what factors influenced the study such as sample size, time etc. It is not a section for excuses as to why your project may or may not have worked.

1.4 Plan of development

Here you tell the reader how your report has been organised and what is included in each chapter.

Factors for architectural choices discussed in lit review Factors for component level design choices discussed in design

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Chapter 2

Literature Review

2.1 Wind Measurement in the Wind Energy Sector

2.1.1 Introduction

Wind has become a mainstream source of electricity. A record 165 GW of new wind capacity was installed worldwide in 2025, 40% more than the previous record year, bringing cumulative capacity to roughly 1.3 TW across 138 countries [10]. Delivering that growth at acceptable cost depends on how well the wind resource itself can be characterised, which Veers et al. identify as one of the grand challenges of wind energy science [24]. Wind speed is an important factor in analysis of a turbine's economics and engineering. The power available in the wind scales with the cube of wind speed, so a small error in measured or predicted wind speed becomes a much larger error in predicted energy [5]. Furthermore, wind speed and turbulence significantly affect the structural fatigue on a turbine, and act as an input to controllers which control the pitch and yaw of the turbines appropriately to maximise power generation and minimise stresses on the structure. It is clear that real-time measurement of the wind field around wind-turbines is a critical objective for this industry, and one to which much attention must be given. However, turbines operate in the lower atmospheric boundary layer, where the flow is turbulent and shaped by mesoscale circulation, orography, thermal stratification and the wakes of neighbouring turbines, which makes this characterisation of the wind-field difficult [18].

2.1.2 Current In-Situ Anemometry Methods and Limitations

The traditional instrument for wind-speed measurement for wind-turbines is the meteorological mast (met-mast), which holds cup and sonic anemometers at fixed heights, mounted on a long mast. This technology is well established, providing high resolution measurements, and indeed acting as a reference point to which any alternative method is compared [19], [13]. This methodology however holds several limitations:

Erecting a meteorological mast can be costly, particularly for offshore wind-turbines [26]. Once constructed, the met-mast is naturally limited to a single location, which limits spatial coverage - this makes variability in the wind, which occurs over larger areas, difficult to characterise [18]. Furthermore, the wake of the mast itself can corrupt measurements in the anemometer [18]. These constraints motivated the research into a remote sensing alternative to in-situ measurements.

2.2 LiDAR as an Alternative

2.2.1 Introduction

Advancements in LiDAR technology have resulted in the technology presenting an alternative and/or complement to existing wind-speed measurement techniques. This allows for the development of new turbine control and assessment strategies, which overcome some of the limitations of traditional wind-speed measurement techniques.

Doppler wind LiDAR (DWL) systems measure the line-of-sight velocity of wind by emitting a laser beam into the atmosphere and measuring the Doppler shift of the beam after being backscattered by naturally occurring aerosols and particles (see section 2.3). There are numerous commercial DWL systems available, such as Qinetiq's ZephIR whose first deployment was on the nacelle of a 2.3 MW turbine, and with Vaisala's (formerly Leosphere) pulsed WindCube shortly afterwards [13] [6]. Research has been focused on two areas; ground-based systems for wind-speed measurement (primarily for resource assessment prior to building wind-energy farms), and nacelle-mounted LiDAR (NML) systems [18].

Letizia et al. identified several promising application areas for NML systems, including load reduction, performance optimization, and performance assessment [18]. NML systems present several advantages over traditional measurement techniques for these ends, including larger field of view and spatial coverage (particularly for scanning LiDARs) and

reduced costs and maintenance [18]. The following subsections provide a brief overview of different NML application areas, and progress and limitations in these sectors. These areas of research present viable means to lower the levelized cost of energy (LCOE) for wind-turbines, and are thus well-suited to the growing demands for wind-energy systems globally [18].

2.2.2 NML for Load Reduction

Wind gusts can place significant load on a wind-turbine structure - however appropriate control of the turbine's rotor pitch can serve to decrease this load, preserving the structural integrity and longevity of the system. Standard turbine controllers are feedback systems that react to a turbulent gust only after it has loaded the rotor. A lidar preview of the approaching wind-field however would enable feed-forward pitch action **before** the gust arrives, and there is consensus that load reduction in certain regions of wind-turbine operation is a promising lidar-assisted control (LAC) application [18]. Crucially however, any benefit from feedforward LAC for load-reduction is limited by the coherence between the wind-speed measured by the LiDAR from a distance, and the actual wind-speed which hits the turbine rotors (rotor-effective wind-speed) [18]. The seminal simulation study of Harris et al. reported a 10% reduction in damage-equivalent load on the flapwise blade root [12], which spurred on further research into the topic. There is much ongoing research on the topic, particularly with respect to the evolution of the wind-field in between sensing and reaching the turbine, and reported results span a wide range. This lack of consolidation has thus presented a barrier to widespread adoption of LAC for load reduction, however research thus far indicates that this is indeed a promising application of the technology [23] [18].

2.2.3 NML for Performance Optimization

Another application area for LAC is performance optimization, where feedforward control based on NML measurements of the incoming wind-field can be used to yaw and pitch the turbine appropriately, and use torque control to maintain optimal rotor speed, so as to maximise power generation. This has however showed limited results in comparison to load reduction. Simulation studies thus far have found negligible gains in power coefficient from feed-forward torque control for optimising rotor speed, because conventional non-LAC torque controllers are already mature [18]. Where gains are reported they come from yaw correction - a hub-mounted ZephIR used to correct misalignment on a 2 MW turbine gave an estimated 5% annual energy production increase [18].

2.2.4 LiDAR for Performance Assessment

Power performance testing is an area in which NML has had greatest success and widespread adoption [18]. A turbine's power curve is conventionally measured against a met-mast's wind-speed measurements. Mikkelsen et al. used a spinner lidar (mounted on the wind turbine blades rather than the nacelle) to reconstruct the low-speed end of a 2.3 MW turbine's power curve and found less scatter than the met-mast measurement, which they attributed to a more representative rotor-equivalent wind speed [20]. Wagner et al. achieved similar results, and identified the reasons: because the lidar yaws with the turbine it always measures upwind, so its measured wind-speed correlates better with power fluctuations than the met-mast's; and because it is never in the wake of the turbine under test, the usable wind sector was larger than the undisturbed sector of the cup anemometer [25]. This shows a direct advantage of a DWL system over traditional in-situ measurements, where the met-mast is fixed. A subsequent uncertainty analysis found the NML estimated annual energy production only 1.1% more uncertain than the cup-anemometer reference, its higher instrumental uncertainty offset by greater data availability [26]. The International Electrotechnical Commission (IEC) formalised the method in 2022 [14], and some 500 campaigns per year are projected this decade [18].

2.2.5 Discussion of NML Applications

The three application areas discussed above present results which show some of the use-cases of LiDAR for wind-speed measurement, however they are not exhaustive, and the topic remains an active area of research. What is however clear in the literature is that the widespread adoption of such systems depends on the reliability, accuracy and cost of the LiDAR systems [18]. Traditionally, such systems have been very expensive, and there has been research into methods to reduce the cost [4]. The remainder of this report thus addresses aspects related to the design of such systems, and the costs and tradeoffs involved.

2.3 Principles of Doppler Wind LiDAR

Doppler wind LiDAR uses a simple working principle in order to determine wind-speed from a distance. The following section details the physics and mathematical concepts which are used in DWL systems.

2.3.1 Aerosol Backscatter and the Doppler Shift

LiDAR systems emit very high-frequency electromagnetic radiation (when compared to its technological cousin - radar), thus, wavelengths of the emitted radiation will be very small. This property makes the technology uniquely suitable for the purposes of wind-speed measurement; the wavelength of the emitted LiDAR beam is comparable to the size of the aerosol particles, and thus the particles scatter some of the emitted beam (Mie scattering) [20].

The backscattered light can be detected by the LiDAR system's photodetector. The backscattered light, however, will be shifted in frequency relative to the incident LiDAR beam, by a magnitude which depends on the velocity of the particle itself - this is due to the Doppler effect [9]. Specifically, the shift in frequency of the backscattered light is directly proportional to the line-of-sight (LOS) velocity of the particle - that is, the component of the particle's velocity in the direction of the line of the LiDAR beam - through the following equation [18]:

$$v_{LOS} = \frac{c}{2}\frac{f_{beam} - f_{scatter}}{f_{scatter}} = \tag{2.1}$$

Since the particle velocities of interest are many orders of magnitude smaller than the speed of light ($v_{LOS} \ll c$), the frequency shift is tiny compared with the optical frequency, and so $f_{scatter} \approx f_{beam}$ in the denominator. Defining the Doppler shift as $\Delta f = f_{beam} - f_{scatter}$ and using $c = \lambda f_{beam}$, where λ is the wavelength of the LiDAR beam, Equation 2.1 simplifies to

$$v_{LOS} = \frac{\lambda}{2}\Delta f \qquad \Longleftrightarrow \qquad \Delta f = \frac{2\, v_{LOS}}{\lambda}. \tag{2.2}$$

The Doppler shift is therefore directly proportional to the LOS velocity and inversely proportional to the wavelength. The factor of two arises because the light is shifted twice: once when it reaches the moving particle, and again when it is scattered back towards the receiver. For the 1550 nm wavelength used in this work (discussed in section 3.4), each m/s of LOS velocity corresponds to a Doppler shift of approximately 1.29 MHz.

Thus, by determining the Doppler frequency shift of the received backscatter from aerosol particles, the LOS velocity of those particles can be determined, and from this, the LOS wind-speed can be inferred. Note that this explanation simplifies concepts used in ranging of the wind-speed, which will be discussed in later sections.

2.3.2 Challenges in Measuring Doppler Shift in DWL

A crucial challenge in DWL is the measurement of the Doppler shift itself. LiDAR beam frequencies are of the order of 10^{14} Hz (hundreds of terahertz) while, from equation 2.1, the Doppler shift in this THz-range signal, produced by aerosols moving at typical wind speeds is of the order of only 10^6–10^7 Hz (a few megahertz) - this is a fractional frequency shift of roughly one part in ten million which needs to be resolved. Photodetectors can not follow typical LiDAR optical signals, which are in the order of THz. For example, for a LiDAR system with beam wavelength 1550 nm, the field oscillates at 193 THz, orders of magnitude beyond the bandwidth of photodetectors or typical ADCs [1, 13]. Thus, the return backscatter signal's frequency can **not** be directly measured, and its frequency relative to the emitted beam simply observed from this. Instead, specialised techniques need to be used to recover the Doppler shift of the signal, **without** having to measure the frequency of the THz-range return signal for comparison. This is thus a central difficulty in DWL systems, and the detection architectures used to address it are discussed in the sections that follow.

2.3.3 LiDAR Detection Modes

LiDAR systems are typically grouped into two main types, each with their own distinct advantages, based on how received light is detected - these types are Direct Detection, and Coherent Detection. To varying extents, both of these detection modes have been applied in DWL systems, with significant differences in how the Doppler shift is measured in each. This section gives an overview of the principles of each detection mode, and discusses how this can be utilised to measure the Doppler shift of received backscatter light.

Direct Detection

Direct detection LiDAR systems operate on a very simple principle: A laser pulse is fired from the system; this pulse then gets reflected off a target (or backscattered in the case of aerosols), and the reflection is incident **directly** on the LiDAR system's photodetector. Owing to its simplicity, direct detection LiDAR systems are widely used in robotics and automotive applications for ranging and basic velocity measurements. The time between the emitted pulse and received pulse can be used to calculate the distance of targets from the system, and repeated distance measurements can be used to estimate the velocity of the target. Obtaining velocity through Doppler shift measurements however presents a

challenge to this architecture.

As discussed in Section 2.3.2, photodetectors by themselves can not be directly used to determine the Doppler shift, as they can not follow THz frequency light signals. Multiple modified direct detection DWL system architectures have been developed to address this problem. Most common is the "double-edge technique" illustrated by Korb et al. [17]. Two Fabry-Pérot Interferometers (etalons) tuned symmetrically around the transmit laser frequency are used to split the backscatter signals - and thereafter the Doppler shift may be obtained from the ratio between the intensity of the two signals, as measured by photodetectors [8].

Crucially for this method, the ratio between scattering types (Rayleigh and Mie) for the aerosols, as well as temperature must be known in order to produce accurate measurements - this can vary significantly with altitude [27]. These factors, and strict requirements on etalon tuning, temperature, and laser stability present a considerable challenge in the implementation of this architecture [8]. In the double-edge technique any drift of the laser relative to the etalon edge is indistinguishable from a Doppler shift, so the transmit laser must be absolutely stable to roughly 100KHz of its nominal frequency, which in practice requires an active lock to an atomic transition [8]. The etalon itself must be temperature controlled to about 1 milli-Kelvin, since a 1 milli-Kelvin drift produces a $0.95m/s$ error, and mechanical drift of the mount can produce biases of $1.5 - 2m/s$ per week, requiring weekly recalibration [8].

An advantage that direct detection architectures hold over coherent detection is the ability to use molecular backscattering return (Rayleigh scattering), which gives it better performance in high-altitude regions with less aerosols to give particulate backscattering (Mie scattering, which a coherent detection system relies on) [8]. However, direct detection DWL systems typically have worse accuracy and spatial resolution than their coherent detection counterparts, which are more technologically mature [27]. For these reasons, Direct detection architectures are not as common in the literature on DWL, as compared to coherent detection systems, particularly in typical wind-turbine applications in the aerosol-rich boundary layer, where the primary advantage of direct detection architectures is unnecessary. The remainder of this review will thus focus on coherent detection systems, and the intricacies of their system architecture.

Coherent Detection

Coherent detection LiDAR varies from direct detection in that the return signal (from backscatter or reflections) is optically mixed with a local oscillator (LO) signal prior to

detection. Typically, a single laser source is used, where a set fraction of the beam is emitted at the target, and the other fraction is used as the LO for mixing down the line. Coherent detection provides multiple distinct advantages over direct detection, which will be described in this section.

Firstly, the optical mixing of the signals makes it possible to electronically distinguish a MHz-range Doppler shift on top of a THz-range carrier (as was identified to be a key challenge in DWL in Section 2.3.2). This can be described mathematically as follows: If the laser source (and therefore LO) has frequency f_0, and the return signal has frequency $(f_0 + \Delta f)$, where Δf is the frequency shift of the return signal due to the Doppler effect, the signal after mixing the return signal with LO is given by:

$$E_{mixed}(t) = \sqrt{2P_{LO}}\cos(2\pi f_0 t) + \sqrt{2P_{RS}}\cos(2\pi(f_0 + \Delta f)t) \tag{2.3}$$

Where P_{LO} and P_{RS} are the time-averaged optical powers of the local oscillator and return signal respectively.

The photodetector current, I_{det}, is proportional to the intensity of the electromagnetic field of the light signal - $|E_{mixed}|^2$. When expanded, one obtains:

$$\begin{aligned} I_{det}(t) \propto\ & \underbrace{P_{LO} + P_{RS}}_{\text{DC}} + \underbrace{2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta f\, t)}_{\text{beat}} \\ & + \underbrace{P_{LO}\cos(4\pi f_0 t) + P_{RS}\cos(4\pi(f_0 + \Delta f)t) + 2\sqrt{P_{LO}P_{RS}}\cos(2\pi(2f_0 + \Delta f)t)}_{\text{not followed by detector}} \end{aligned} \tag{2.4}$$

It can be seen from the above equation that the squared mixed signal has three components: one at DC, one with frequency in the order of f_0, and another with frequency at Δf. Note again that f_0 is in the THz range, where Δf is in the MHz range. As discussed, if a MHz-range bandwidth photodetector is used to detect E_{mixed}, the detector cannot follow oscillations at frequencies in the order of f_0, so those terms produce no response and may be omitted. Thus, Equation 2.4 simplifies to the following equation, which is called the beat signal:

$$I_{det}(t) \propto [P_{LO} + P_{RS}] + 2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta f\, t) \tag{2.5}$$

Thus, if the frequency of this beat signal is measured, the Doppler shift, Δf, is obtained.

In this way it can be seen that coherent detection allows for direct measurement of the Doppler shift, and therefore the velocity of the target.

Aside from the direct measurement of the Doppler shift, the "signal beating" process described above provides several other key advantages. The beating process amplifies the return beam, which allows for shot-noise limited operation (the ideal case for such a system), and extremely high sensitivies [13] (elaboration in 2.3.6). This is essential in DWL, as return signals in clear atmospheric conditions are typically very low, due to low backscatter from aerosols. Furthermore, this detection process rejects signals which are not common in spatial mode and polarisation with the LO, meaning that sunlight, or other LiDAR beams will not affect the measurement [13].

2.3.4 Coherent Detection: Homodyne, Heterodyne and Directional Ambiguity

The following section discusses two distinct classes of coherent detection methodology, and how the choice of class affects the recovery of the wind direction from the signal.

Homodyne

While not explicitly stated, the previously discussed direct down-conversion of the return signal to the beat signal - via mixing with a local oscillator with the **same** frequency as the emitted signal (Equation 2.5 and preceding) - is known as **homodyne** coherent detection. Homodyne coherent detection down-converts the MHz-range doppler shift in the THz range carrier to a MHz-range beat signal around baseband, i.e. the frequency of the signal is directly the frequency of the Doppler shift. This has significant implications:

Ignoring the DC components in Equation 2.5, we get

$$I_{det}(t) \propto \cos(2\pi\,\Delta f\,t)\,. \tag{2.6}$$

The Doppler shift (Δf) therefore appears directly at baseband. However, because the cosine is an even function,

$$\cos(-2\pi\,\Delta f\,t) = \cos(2\pi\,\Delta f\,t)\,, \tag{2.7}$$

Therefore, noting that Δf is signed based on whether the target is approaching or receding, it can be seen that approaching and receding targets of equal speed produce identical detector currents. Only the magnitude $|\Delta f_D|$ is recoverable, and the measured spectrum

is symmetric about zero frequency. Note that Abari et. al[1] demonstrated a methodology for obtaining signed velocity, even with a homodyne coherent detection setup, however this setup is novel and introduces additional complexities and costs which are outside of the scope of this project.

Heterodyne

The alternative to this is known as **heterodyne** coherent detection. In heterodyne architectures, a slight frequency shift is imposed on the laser - crucially however this shift is only applied to one of the fractional outputs, either the local oscillator **or** the emitted laser (but not both). This means that when the same beating process described previously is applied, the MHz-range doppler shift in the THz range carrier is down-converted to a MHz-range beat signal centered around an intermediate frequency (f_i below), rather than baseband, as with homodyne architectures. This is conventionally achieved with an acousto-optic modulator, placed either in the local oscillator arm [11] or in the transmit arm [3]. This has the advantage of allowing for the recovery of the target's direction:

Taking the offset in the local oscillator arm, such that $f_{LO} = f_0 - f_i$, the beat term becomes

$$I_{det}(t) \propto \cos(2\pi\,(f_i + \Delta f)\,t)\,. \tag{2.8}$$

Provided $f_i > |\Delta f|_{\max}$, the sign of Δf is recoverable. Approaching targets appear above f_i and receding targets below it, and the sign of the velocity is recovered.

2.3.5 Coherent Detection: Noise Sources and the Shot-Noise Limit

In a coherent detection DWL system, there are four primary sources of noise which need to be considered, namely electronic noise, shot noise, relative-intensity noise, and phase-induced intensity noise:

Electronic noise arises from the photodetector and the transimpedance amplifier (which converts photodetector current into a measurable voltage), and includes dark noise (signal present when there is no light incident on the detector) and thermal noise [13]. This noise is constant, independent of LO power [6], [3].

Shot noise is a fundamental noise source in coherent detection. Shot noise arises from the discrete nature of photons, which reach the photodetector at slightly different random instants [21], i.e. it is a property of light itself, and can therefore not be removed. Because

the photodetector's current is generated by these received photons, the detector then produces a white, Gaussian noise current whose power spectral density is proportional to the intensity of the light incident on it [21]. In the case of a coherent detection setup, it is thus proportional to P_{LO}. The significance of shot noise follows from comparing its scaling with that of the beat signal. From Equation 2.5 the beat current has amplitude $\propto 2\sqrt{P_{LO}P_{RS}}$, so the signal power is proportional to P_{LO}. Shot noise power is also proportional to P_{LO} [7, 3]. Electronic noise power, however, as mentioned, is constant with P_{LO}. Neglecting other noise sources for the moment, the SNR is approximately [7]:

$$\text{SNR} = \frac{P_{LO}P_{RS}}{P_{EN} + P_{SN}} = \frac{P_{LO}P_{RS}}{P_{EN} + \alpha P_{LO}} \tag{2.9}$$

Where P_{EN} is electronic noise power, P_{SN} is the shot noise power, and α is a proportionality constant representing the other terms which multiply with P_{LO} to give the shot-noise power. Observing thus that P_{SN} grows proportionally to P_{LO}, it can be seen that at low LO powers the electronic noise term dominates, and the SNR will grow as P_{LO} scales (because the numerator scales with P_{LO} as well). At very large LO powers, the electronic noise power term becomes negligible relative to the shot noise, and thus the SNR simplifies to:

$$\text{SNR} \approx \frac{P_{LO}P_{RS}}{P_{SN}} = \frac{P_{LO}P_{RS}}{\alpha P_{LO}} = \frac{P_{RS}}{\alpha} \tag{2.10}$$

This is called ideal **shot-noise limited** operation, and it is the best possible operating point for a coherent detection system - the LO has raised the signal (and shot noise) so far above the electronic noise floor that the electronic noise floor is swamped out so as to become negligible, and no further sensitivity is available from the receiver. The SNR reaches a constant peak which is proportional only to the received backscatter signal power (P_{RS}), and the constant α which governs shot-noise power. [4]. In practice however, the ideal shot-limited operating point can not be fully reached, because two effects prevent P_{LO} from simply being made as large as possible: the photodetector saturates [3], and the laser's intensity noise, discussed next, grows faster than the signal.

Relative-intensity noise (RIN) is fluctuation of the source laser's relative intensity, which appears as current noise at the output of the detector [13], creating a wideband noise floor in the frequency domain. As can be derived from Equation 2.5, the useful detectable signal **power** grows with P_{LO}. However, RIN power grows with P_{LO}^2, and thus grows faster than signal power [3]. Recall from above that we desire a large P_{LO} in order to achieve shot-noise limited operation, with larger P_{LO} giving a better SNR in Equation 2.9 **when RIN is not included**, and the ideal case being an infinitely large P_{LO}. With RIN

included however, as in:

$$\text{SNR} = \frac{P_{LO}P_{RS}}{P_{EN} + P_{SN} + P_{RIN}} \tag{2.11}$$

It can be seen that beyond a certain P_{LO} level RIN will overtake shot noise in the denominator, and the system's SNR will no longer be increasing with increasing P_{LO} and approaching the ideal shot-limited regime [3]. Instead, the SNR will **decrease** with an increasing P_{LO} after this crossover point, due to the P_{LO}^2 scaling of RIN [7]. It can thus be seen that RIN establishes an optimal P_{LO} which can be used, and sets a ceiling on how close the detector can get to the ideal shot-noise limited operation of Equation 2.10, before increasing RIN dominates. It is therefore clear that if RIN can be reduced through other means, a larger P_{LO} can be used, and thus the system can get closer to the ideal shot-noise limited state. Balanced detection is one means to achieve this [7], and is discussed in subsection 2.3.6.

Finally, there is phase-induced intensity noise (PIIN), which is noise which results from non-ideal linewidth effects of the laser source [7], [15], [11]. PIIN is a significant source of noise in DWL systems, and has been shown to be highly unstable, and to possibly exceed the shot-noise by up to 30dB [11]. Thus, in order to remain as close as possible to the ideal shot-noise limit, PIIN should be reduced. This is possible through the selection of an appropriate laser source, and through careful matching of signal path lengths in the DWL system [11][7]. PIIN is a complex noise source which can vary significantly in shape and magnitude depending on the laser source and the DWL system. An understanding of laser phase noise and linewidth is required to understand it - a full discussion and mathematical breakdown on these concepts and how they relate to PIIN is provided in section A.3.

2.3.6 Coherent Detection: Single-Ended vs Balanced Detection

As previously discussed, in order to determine the Doppler shift of the received backscatter signal in a coherent detection DWL system, the backscatter signal must be optically mixed with the LO signal, and this mixed signal must then be detected using a photodetector system. This section discusses the two common types of photodetection methodologies - single-ended detection, and balanced detection. Balanced detection holds two key advantages over single-ended detection, which will be discussed here.

A single-ended detection system simply uses a 2:1 (2 inputs, one output) fiber optic coupler to mix the LO reference signal with the backscatter signal, and uses a single

photodetector to convert this light signal into a measurable current. Balanced detection instead makes use of a 2:2 fiber optic coupler for mixing, thus giving two outputs which will be opposite in phase (due to conservation of energy [22]). These opposite phase signals are then detected using two matched photodiodes, with the output being given by the differential of the current produced by the two. At this point it is important to note that a 2:1 coupler functions exactly the same as a 2:2 coupler, but with one of its outputs terminated - the effect of this being that effectively half of the signal is discarded. In a balanced setup the two antiphase beat signals are subtracted, so their amplitudes add, doubling the beat current amplitude (a fourfold increase in signal power) relative to a single 50:50 output [16]. Because the shot noise of the two photodiodes is **uncorrelated**, the noise power only doubles, giving a net 3dB improvement in shot-noise-limited SNR over single-ended detection with a 50:50 coupler [22]. Conversely, RIN is common to both outputs and is **cancelled** by the subtraction [2, **?**]. This allows the system to better approach the ideal shot-noise limit sensitivity as discussed in subsection 2.3.5. For a mathematical description of why this is the case, and a full treatment on the significance, see A.1.

2.3.7 Transmit and Receive Optics: Monostatic vs Bistatic

For a DWL system there are two optical configurations which are commonly used - bistatic and monostatic. This section discusses the differences between the two.

Bistatic

Bistatic configurations use one lens to emit the laser into the atmosphere, and another to receive the backscatter signal. One distinct advantage of a bistatic configuration, is that back reflections of the beam from the lens used for emission are not coupled into the mixed signal, and thus do not appear at the detector [7]. This serves to reduce phase induced intensity noise (discussed in section 3.4.1. However, this is at the cost of greatly increased complexity, due to additional considerations of alignment of the two lens apertures [11] [4].

Monostatic

A monostatic configuration uses a single lens aperture for both the emission of the laser, and the receiving of the backscatter signal. A fiber optic circulator can then be used to ensure the emitted signal and the received signal do not interfere, and are routed appro-

priately in the system. This has the distinct advantage of not requiring any additional lenses, and alignment considerations, and indeed has become the norm for such systems due to this reduced complexity [13].

2.3.8 Ranging: Pulsed versus Continuous Wave

Coherent DWL systems have two distinct methodologies which can be applied in order to determine at what distance from the system a measured wind-speed is occurring. The methodology differs depending on a key architectural choice of the system, namely whether the LiDAR beam is on for longer periods of time - known as continuous wave (CW) - or if it is pulsed. Both of these architectures hold their own advantages and disadvantages, which will be discussed in this section.

Pulsed LiDAR

As the name suggests, a pulsed LiDAR system emits its beam in a short pulse. Depending on the pulse duration, a given volume of air - called the probe volume - will be illuminated at an instant of time, and will create backscatter. This volume naturally moves through the air as the beam propagates radially. Thus, by using extremely precise timing on the returned backscatter, it can be determined which distance "slice" of air that backscatter was generated by, and thus, what the wind-speed for that particular "slice" is. The wind-speed for that distance is averaged over the probe volume, thus a shorter pulse will give greater spatial accuracy due to its shorter probe volume. This technique is known as range-gating, and has been widely applied in commercial and scientific DWL systems. [6] [25] [18]. A distinct advantage of range-gating is that the probe volume remains the same at all distances, including longer ones, giving it constant range resolution for longer distances than CW systems [6] (described below). However, it must also be noted that typical pulsed systems are ineffective in ranges below tens of metres, because the pulse length is at least as long as the desired range [13].

Continuous-Wave LiDAR

Continuous-wave systems emit a continuous beam through the atmosphere. This simplifies the design cost and complexity of the systems [6] [13], and no precise pulse timing or receive timing is required. CW systems achieve ranging by using optics to focus the beam at a specific distance. This gives a Lorentzian spatial weighting function along the beam length, with a peak at the focus point of the beam (the point with highest intensity), thus

concentrating the detection sensitivity around the focus, at the expense of contributions from other ranges. However, the effective probe volume varies **quartically** with the distance of the focal point from the source, thus significantly increasing the volume over which the Doppler shift (and therefore wind-speed) is averaged at longer focal distances. Therefore for longer ranges, spatial resolution severely decreases, and it is not possible to gate the measurements for any specific region of air before the focal point. For this reason, CW systems are not well-suited for long ranges of several hundred meters [13], however at short ranges they can be more effective than pulsed systems, owing to the previously mentioned limitations of pulsed systems, relating to pulse length.

Chapter 3

System Design

3.1 Introduction

This chapter details the design of a Doppler wind LiDAR system. As discussed in the introduction and scope of this report, the system is a benchtop prototype whose purpose is to make the sensitivity of a DWL measurable and attributable to specific component and architecture choices. As such, the primary considerations will be noise-levels, sensitivity, and the overarching principles of physics governing DWL systems. This system design will thus not include the design of housing for field deployment, nor will it be concerned with real-time post-processing pipelines for the system's data (which would be required in a field-deployed system). Furthermore, several key components that would be typical in a scientific- or industry-grade, field-deployable DWL system are deliberately omitted in order to reduce costs, at little functional expense for a benchtop system. Where possible, the system was made extensible for future iterations and field deployment.

First subsection 3.2.1 gives an overview of the architecture chosen, as synthesised from the various architecture types reviewed in chapter 2, and with respect to the project's overarching goals. Thereafter, section 3.3 discusses the functional and performance requirements of the system. Finally, section 3.4 details the specific components chosen to meet these requirements.

3.2 System Architecture

From the reviewed literature in chapter 2, it was decided that an **all-fibre, coherent detection, continuous-wave, monostatic** DWL system with **balanced detection**

would be built, with sampling and digitisation to be handled by a benchtop oscilloscope, and post-processing done on a laptop computer. This section details why these architectural choices were made, using summarised information from chapter 2 with respect to the project goals, and discusses any significant implications for the system.

3.2.1 Coherent detection

The direct detection technique requires the Rayleigh/Mie scattering ratio and temperature, an absolutely stabilised laser, etalon control to 1 milli-Kelvin, and very precise physical alignment and calibration [17, 8, 27]. These present substantial challenges to an undergraduate project, particularly the stabilised laser and etalon temperature control. The one advantage which direct detection holds, which is the capability of using molecular backscatter when aerosol backscatter is not available, is irrelevant in an aerosol-rich controlled lab environment, which is the only environment this system will be tested in.

By contrast, coherent detection is far more common in the literature, and thus has more practical information and sources for components. Furthermore, mixing with the LO also permits shot-noise-limited operation through the effective amplification of the return signal, and rejects unwanted light not mode-matched to the LO [13] (e.g. other LiDAR systems, or the sun), thus reducing sources of potential interference. The architecture thus lends itself to a more elegant, direct method of measuring the Doppler shift, via optical mixing and subsequent detection.

For these reasons, a coherent detection system was chosen for this project build.

3.2.2 Homodyne mixing

Homodyne mixing places the Doppler shift at baseband with no additional frequency shift, and no additional components necessary, whereas heterodyne operation requires an acousto-optic modulator (AOM) in the LO or transmit arm [11, 3]. The tradeoff is the sign ambiguity as discussed in subsection 2.3.4, where only the velocity **magnitude** is measured in a homodyne system, whereas a heterodyne system can resolve the direction.

For a benchtop-scale system which primarily aims to present the working principles of a DWL system and a sensitivity analysis, and where the direction of the flow being measured will always be known, it is not necessary to recover signed velocity. Adding an AOM thus adds needless complexity, and adds an additional unnecessary cost to a budget-constrained project. Crucially however, the addition of an AOM for a future iteration of the system

is simple, and does not require the redesign of any parts of the current system. For these reasons it was thus decided to omit the AOM, and to therefore construct a **homodyne** coherent detection system.

3.2.3 Balanced detection

As discussed in subsection 2.3.6, balanced detection holds the advantage of providing a 3dB improvement in SNR (in shot-noise limited operation) as compared to single ended detection. It also allows RIN to be largely cancelled, further allowing SNR improvement, and allowing the system to better approach the shot-noise limit (see Appendix A.1). For these reasons, it was decided that a balanced detection setup would be used.

3.2.4 Monostatic optics

For this project, it was decided to use a monostatic optical configuration - where a single aperture serves both transmit and receive, separated by a fibre circulator, with no second lens and no alignment of two lenses necessary. A bistatic arrangement would keep transmit-optics back-reflections away from the detector, at the cost of greatly increased alignment complexity [7, 11, 4], as well as the additional cost of 2 sets of optics, and equipment needed for alignment. The accepted tradeoff here is thus that back-reflections from the transmit path reach the detector and raise the noise floor through PIIN (see section 3.4.1).

3.2.5 Continuous-wave operation

Pulsed systems are ineffective below tens of metres because the pulse length is typically at least as long as this range [13]. a CW system needs no pulse or receive timing, which simplifies design and can reduce costs (due to lower component count). It is also the more effective of the two at short range. One of the significant performance costs of a CW system however is reduced spatial resolution, particularly at long ranges, as spatial resolution is defined by focus rather than gating with timing (see **??**). Seeing as a benchtop system requires operation at short ranges in the order of tens of centimeters to one or two meters, a CW system will be best suited here due to its simplicity, and its advantages at short range, as well as the irrelevance of long-range limitations.

3.2.6 All-fibre construction

For this project it was decided that an all-fibre construction would be used - meaning that fibre optic cables would be used to transmit the LO and emission beams, and return signals, rather than transmitting the beams through free-space. Single-mode fibre makes the mode-matching between LO and return that coherent detection requires automatic, removes free-space alignment complexities, and gives access to widely and cheaply available telecom components [4, 1, 3]. Non polarisation-maintaining fibre is used, owing to cost considerations as compared to polarisation-maintaining fibre. The accepted tradeoff of standard (non-polarisation-maintaining) fibre is that the polarisation of LO and return is not fixed, which will reduce the heterodyne efficiency of the system, and thus reduce the measureable Doppler peak [7].

3.2.7 System Architecture Diagram

Figure 3.1 below shows a functional block-diagram of the system's decided-upon architecture. In this diagram, the signal path can clearly be seen from laser source, through the splitter (splitting into LO and transmit beam). Thereafter the transmit beam gets routed via port 1 –>2 of the circulator to the monostatic optics, which also receives the backscatter signal to be routed via port 2 –>3 of the circulator. The local oscillator and backscatter signals are mixed in the optical mixer, and then detected by a balanced photodetector, from which the final output beat signal can be sampled and processed.

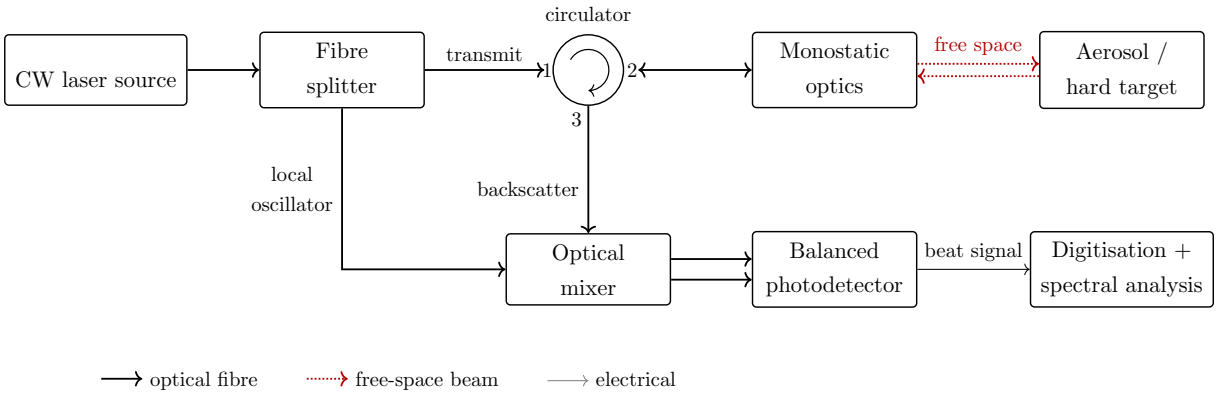


Figure 3.1: Block diagram of the chosen architecture: all-fibre, coherent, continuous-wave, monostatic, homodyne, with balanced detection

3.3 System Requirements Analysis

Now that a system architecture has been decided on, requirements which are specific to this choice of architecture, and which are related to the project goals can be synthesised. The requirements for the system are divided into two sets: functional requirements, which define the ideal outcomes which are desired from the system, and performance requirements, which define specific metrics which are to be measured and quantified with respect to the goals of the functional requirements.

3.3.1 Functional requirements

1. Measure the line-of-sight velocity of a target from the Doppler shift of coherently detected backscatter, using a monostatic all-fibre homodyne architecture with balanced detection and continuous-wave operation.
2. Operate on both a diffuse hard target of known velocity and on air seeded with smoke or fog.
3. Produce a velocity estimate from a recorded spectrum using a stated, reproducible detection criterion (threshold relative to the system's noise floor).

3.3.2 Performance requirements

From the desired functional requirements, a set of performance requirements can be synthesised; these are given in Table 3.1 below along with the IDs of the tests which will verify them.

ID	Requirement	Value	Verified by
P2	Detector noise	approaching ideal shot-noise-limited behaviour: total detector noise (electronic noise + residual RIN after balanced detection) within 1 dB of the shot-noise limit	A1, A2
P4	Phase-induced intensity noise (PIIN)	Level (in dB) and spectral shape measured; agreement with the path-mismatch model of [11] within 6 dB	B
P3	Total system noise	Total noise floor (shot + electronic + residual RIN + PIIN) quantified with respect to the shot-noise limit, and sources of excess noise described and characterised	B
P1	Velocity detection on a hard target	Doppler peak detected at $\gg 5 \times \sigma$ (5 times the noise standard deviation) above the mean, at $N \leq 1000$ averages, for a hard target 10cm away from laser emission	C
P5	Low-velocity limit	Minimum resolvable LOS speed of $\leq 0.1\,\mathrm{m\,s^{-1}}$ on a hard target	C
P6	Detection of wind-speed in seeded air (smoke or fog)	Doppler peak detected at $\gg 5 \times \sigma$ (5 times the noise standard deviation) above the mean, at $N \leq 10000$ averages	D
P7	Velocity accuracy	Doppler velocity within 5% of ground-truth velocity of target (hard target and seeded flow)	C, D

Table 3.1: Performance requirements and their verification tests.

3.4 Component Choices

This section details the specific component selections which were made for the system. These selections were made to best suit the functional and performance requirements of the system. It must be noted that this project is constrained when compared to typical scientific or industrial DWL systems. As such, many component choices were driven by cost, availability, and time-frame considerations, rather than simply their suitability for the system. This is however a suitable approach for the goals of the project, which are to illustrate the principles and physics involved in a DWL system, and to quantify system performance with respect to component choices.

3.4.1 Laser Source

The selection of an appropriate laser source is critical to ensure good performance of a DWL system. There are numerous properties of a laser source which should be considered, and which will be detailed in this section. After discussion on these properties, a suitable laser source is selected.

Wavelength

There are many suitable wavelengths for DWL systems, as detailed by Cariou et. al [7]. There is much discussion around SNR optimisation and backscatter maximisation for varying wavelengths [4], however owing largely to practical advantages, a wavelength of 1550nm has emerged as a common choice for both commercial and scientific DWL systems [1]. Fiber optics telecommunications commonly operate at this wavelength - thus making components designed for this wavelength readily commercially available and significantly cheaper than alternatives [7], [3], [13]. Additionally, laser beams at 1550nm are eye-safe at significantly higher power levels (up to multiple Watts), compared to shorter wavelengths [7], reducing safety concerns. For this reason it was decided that a 1550nm laser source would be used for this project.

Power Level

As previously discussed, the ability of a DWL system to measure wind-speed depends on the measurement of aerosol backscatter of the emitted beam.

The time-averaged power of the backscatter signal into a receiver can be expressed as [13]:

$$P_s = \pi P_t \beta(\pi) \lambda \tag{3.1}$$

Where P_s is the time averaged backscatter signal power, $\beta(\pi)$ is the atmospheric scattering coefficient (evaluated at π, i.e. scattering at π radians, back towards the source), and λ is the wavelength of the transmitted light. Note that $\beta(\pi)$ also holds a wavelength dependence, and thus P_s does not necessarily scale linearly with increasing wavelength, as the equation may suggest, but instead resolves to a near cancellation of the wavelength dependence between some frequency ranges [4].

It can be seen in Equation 3.1 that there is a linear dependence on P_t, the power of the transmitted beam. Thus, a higher-powered transmit laser source would give a larger

return signal to be measured, increasing the sensitivity of the system by increasing the SNR of the system [4].

A setup commonly used to achieve higher emitted laser power here is the master-oscillator power-amplifier (MOPA) configuration [7], [4]. This separates the functions of the laser source - allowing for a low-power, highly coherent, narrow linewidth laser source to be used as a master-oscillator and an amplifier to increase the power levels of the master oscillator beam, while maintaining its spectral purity. This also allows the local oscillator to be derived from the master oscillator, eliminating the need to have two separate laser sources - one for LO, one for output - which would need to have a stable frequency relationship [7], something which requires careful design to achieve, as illustrated by Colberg et al. [8].

As noted by Bennett, such power amplifiers (commonly Erbium-doped fiber amplifiers; EDFAs) make up a significant portion of the cost of a DWL system. Indeed, more than 20 years after the writing of Bennett's paper, the cost of EDFAs remain prohibitively expensive for a project at this scale. For EDFA systems which are able to amplify signals to hundreds of milli-Watts or multiple Watts (which is a typical power level for such a system [3][6][7]), current prices range in the thousands to tens of thousands of USD (see Appendix).

For this reason, the use of a power amplifier for the laser source is disqualified for this project on the grounds of cost. The system was however designed such that the addition of such an amplifier would be possible at a later stage.

Due to the relatively low powers available for laser sources without power amplification (in the tens of milli-Watts), other avenues to improve detection sensitivity will be required. Bennett identifies another option for increasing the sensitivity of a DWL system - which is to average over many returns [4]. This is the approach which will be taken for this project. Additionally, owing to this constraint in laser source power, all tests which concern wind-speed measurement in air will be performed using air seeded with smoke and/or fog. This will increase backscatter levels by increasing $\beta(\pi)$ in Equation 3.1, and thus reduces the need for the power amplifier for this benchtop-scale system. The level of seeding required for a lower-power laser source to detect wind-speed is one of the investigations of this project.

Linewidth

Linewidth can be seen as a measure of the spectral purity of a laser source, and is one of the most critical parameters in the design of a DWL system. A narrow linewidth is

highly desirable for DWL applications, as this will significantly reduce the phase-induced intensity noise (PIIN) in a DWL system, and allow the system to therefore have a better SNR and sensitivity to detect small backscatter signals. Furthermore, the linewidth of a laser source has a significant effect on how broad the returned backscatter signal's Doppler shifted peak is, particularly at longer ranges. This will result in the system having worse precision, and not being able to resolve turbulence in the air as well. For a full discussion on the mathematical and physical meaning of linewidth, as well as a treatment on how linewidth affects PIIN and the broadened peak, see Appendix A.3. Cariou et al. make the claim that linewidths typical of DFB diodes (0.5 - 5MHz, see Table 3.2) disqualify them for use in CW DWL systems due to their PIIN, however, Bennett shows that indeed a DFB source with a linewidth of 1.4MHz can be used for these purposes, provided careful attention is given to other design considerations, such as optical fiber path length matching [4]. The analysis of noise in the system, and the associated constraints arising from the use of a comparatively wider linewidth laser source indeed forms a key part of this project, owing to budget considerations, as will be discussed below and in **??**.

Laser source types and selection

There are many types of laser sources available on the market. In the selection of a laser source for this project, the primary driver was cost and availability, with the design goals being to maximise power level and minimise linewidth, while fixing the wavelength at 1550nm. After selecting the best available laser source for these parameters, the performance of the laser source will then be analysed, and recommendations based on its properties and performance can then be made.

Source type	**Linewidth (typ.)**	**Power, unamplified**	**Noise and practical notes**	**Indicative cost (USD)**
Fabry–Pérot laser diode	multi-mode; not single-frequency	10 mW to 100 mW	Multi-mode, Unsuitable for coherent detection due to this. [11].	< 100
DFB laser diode	0.5 MHz to 5 MHz	10 mW to 40 mW	Relaxation peak ∼1 GHz, far above the Doppler band, thus is RIN flat and low in the frequency band of interest [7]. Needs external driver and temperature controller.	100 to 1000 (module)
Planar external-cavity semiconductor	$<$ 1 to 15 kHz	10 mW to 30 mW	Integrated driver, marketed for wind lidar.	quote only; low thousands
Er-doped DFB fibre laser	$<$ 0.1 to 10 kHz [11]	10 mW to 40 mW	Common source of choice for CW anemometry due to narrow linewidth [7, 13, 3]. Relaxation peak near 1 MHz, so higher RIN in the frequency band of interest [7].	quote only; tens of thousands

Table 3.2: Laser source classes at 1550 nm. Linewidths and powers are typical of commercial devices; approximate costs are as of September 2026 and exclude drivers, controllers and import costs.

From these available laser source options, it was decided that a DFB laser diode would be used. While the wider linewidth, as compared to Planar external-cavity semiconductor diodes or Er-doped DFB fibre lasers is likely to substantially increase PIIN, measures such as fiber length matching will be investigated to reduce PIIN for lower-range, tabletop operation (see **??**), and these narrow linewidth sources are out of budget for this project.

The final specifications of the obtained laser diode are as follows:

Parameter	Value
Wavelength	1550 nm
Rated output power	20 mW
Linewidth	2 MHz
Operating current at 20mW	58 mA
Fibre output	Single-mode, FC/APC

Table 3.3: Key properties of the selected DFB laser diode (BFLD-1550-20SM-FA2). Full specifications are given in **??**.

3.4.2 Laser driver and temperature control

In order to keep the laser diode at a constant operating point, a constant current source must be used to drive it, and a temperature controller should be used. Temperature and current fluctuations present in the laser output as fluctuations in wavelength and output power [**?**], which can have a significant effect on the noise in measurements, particularly in a sensing system, such as a DWL, for which measurements of very small signals is required.

Perhaps the greater consideration however is the safety of the laser diode. The laser diode chosen forms the single biggest expense in this DWL system, and thus protecting it is of vital importance to this project. Laser diodes are extremely sensitive to transients and overvoltage events, which can easily damage them [**?**]. Circuitry for these ends may be custom-designed and manufactured as a PCB, however, this is a complex process, and is best-suited to systems where commercially available driver and temperature control circuitry is unsuitable. Thus, for this project it was decided that off-the-shelf circuitry would be used for these ends, so that the expensive laser diode is safeguarded against possible custom design flaws or oversights of a first-time designer. Commercial off-the-shelf units integrate soft-start circuitry, current control, current monitoring, protection against overvoltage and overcurrent events and transients into a single package.

The Koheron DRV200S-A-200 laser driver was chosen, due to its very low noise characteristics, and the suitability of the current range and limit for the BFLD-1550-20SM-FA2 laser diode chosen above. The Koheron TEC100L-50 temperature controller was chosen from the same supplier, and has an adequate setpoint range and stability for the project. The respective specifications of the selected laser driver and temperature controller are given below:

Parameter	Value
Current range	0–200 mA
Current limit (adj.)	65–300 mA
Noise density (1 kHz)	270 pA/$\sqrt{\text{Hz}}$
RMS noise (10 Hz–1 MHz)	265 nA
Temperature coefficient	30 ppm/°C

Table 3.4: Key specifications of the Koheron DRV200S-A-200 laser diode driver.

Parameter	Value
TEC current	±1.15 A
Setpoint range	10–40 °C
24 h stability	0.005 °C
Temperature coefficient	0.002 °C/°C
Supply voltage	5 V

Table 3.5: Key specifications of the Koheron TEC100L-50 temperature controller.

3.4.3 Detector

Balanced detectors typically include two matched photodiodes, connected differentially to a low-noise transimpedance amplifier which converts the photodiode current into a measurable voltage. Seeing as the beat signal which will be measured is in the order of mega-Hertz in frequency, much care needs to be taken when designing the electronic circuitry of the detector to avoid parasitic effects, such as stray capacitance and inductance, which can cause amplifier instability and limit bandwidth, as well as susceptibility to electromagnetic interference. Given the difficulty of proper RF design, to simultaneously achieve high bandwidth, low noise and high common-mode rejection in a custom design, a commercial balanced detector was selected so that project effort could be focused on the lidar system itself.

The Koheron PD100B Balanced Photodetector was selected for use in this project, with the key specifications given in Table **??** below:

Parameter	Value
Photodiode responsivity	0.9 A/W
Max. input power (per diode)	1.5 mW
Bandwidth (−3 dB)	160 Hz–100 MHz
Transimpedance gain	39 kV/A
Input current noise density	8 pA/$\sqrt{\text{Hz}}$
CMRR (1 MHz)	35 dB

Table 3.6: Key specifications of the Koheron PD100B-AC balanced photodetector. Full specifications are given in **??**.

The photodetector has a bandwidth of 160Hz - 100MHz, thus DC photocurrent is blocked

(this is desired, seeing as the measured quantity of interest is the frequency power at MHz-range frequencies). This will also allow signals of up to 100MHz to be effectively detected. A beat frequency of 100MHz corresponds to a wind-speed of approximately 129m/s (464km/h), which is clearly far above any wind-speed which the system would reasonably need to detect. It can also be seen that the detector has a very low electronic noise floor of 8pA/$\sqrt{Hz}$, which can easily be swamped out by the shot-noise from the LO to achieve shot-noise limited behaviour, and the CMRR is high, allowing RIN to be very effectively cancelled.

3.4.4 Fiber Optics

Optical Mixer, Circulator

A standard 50:50 non-polarisation maintaining FC/APC connector type fiber optic coupler is to be used as the optical mixer (see Figure 3.1). This has two input signals which are mixed at a 50:50 ratio, and split into two output signals for balanced detection.

A standard 3-port non-polarisation maintaining FC/APC fiber optic circulator is obtained to allow transmission and reception of signals from the same optics.

Fiber Splitter

More care must be taken in selecting the fiber splitter, as the ratio here determines how much of the laser source power is directed towards emission, and how much towards use as a local oscillator for down-stream mixing.

As discussed in chapter 2, the LO power has a significant effect on the noise characteristics and the performance of the system. While the LO power can be indirectly varied by varying the laser source power (via current), this will also vary the transmit power. Thus, a splitter ratio should be selected which allows for a suitable LO power level which allows for shot-noise limited operation, when the laser is set to an output power level which provides sufficient transmit power.

The lower bound on the LO power is set by the requirement for shot-noise-limited operation, which occurs when the LO shot noise exceeds the electronic noise of the detector (see subsection 2.3.5). For the balanced detector, the shot noise current density is $I_{\text{sn}} = \sqrt{2qRP_{\text{LO}}}$ (derived from Kikuchi et al. [16]), where R is the photodetector's responsivity, and q is the charge of an electron, so equating this to the PD100B input noise

density $I_n = 8$ pA/$\sqrt{\text{Hz}}$ gives

$$P_{\text{LO,min}} = \frac{i_n^2}{2qR} = \frac{(8 \times 10^{-12})^2}{2(1.602 \times 10^{-19})(0.9)} \approx 222\ \mu\text{W}. \tag{3.2}$$

The upper bound is set by photodiode saturation. The 50:50 coupler divides the LO equally between the two photodiodes, which are each rated to a maximum of 1.5 mW, so $P_{\text{LO,max}} \approx 3$ mW.

For a splitter that directs a fraction κ of the laser power P_L to the LO, the condition $P_{\text{LO,min}} \leq \kappa P_L \leq P_{\text{LO,max}}$ defines the range of usable laser powers. With $\kappa = 0.1$, this gives 2.2 mW $\leq P_L \leq$ 30 mW, which spans the full output range of the 20 mW source. At full power, this yields an LO of 2 mW, approximately 9× (9.5 dB) above the electronic noise crossover and comfortably below saturation, while 18 mW is directed to transmission. By comparison, a 50:50 splitter would saturate the detector for any $P_L > 6$ mW, limiting the transmit power to 3 mW. A 99:1 splitter would provide only 0.2 mW of LO at full laser power, leaving the system electronic-noise-limited across its entire operating range. The 90:10 ratio was therefore selected, as it keeps the LO power within the shot-noise-limited window across the full range of laser drive currents. This also makes it possible to vary the LO power by adjusting the laser power across its full range, when later investigating the effect of LO power on the system's noise (see **??**).

3.4.5 Optics

A transmit/receive optic is required to emit the laser output into free space and to collect the backscattered light, which the circulator then routes to the detector. Since the system is a continuous-wave system and not pulsed (see **??**), any ranging must come from the focusing of the output beam. However, owing to the scope of the project, which is primarily to investigate the design considerations in a **benchtop** DWL system, with a lower power laser which will only be used in a controlled, seeded air-flow in a wind-tunnel, ranging is not a primary point of investigation for the project.

A collimator was thus selected (Table **??**). This produces a collimated beam of approximately 7 mm diameter, with a Rayleigh range of approximately 25 m - meaning that the diameter does not significantly change over 25m. In a small wind-tunnel which is much shorter than 25m, the beam diameter, and hence the receiver sensitivity, is effectively constant along the beam. The measured Doppler spectrum therefore represents the line-of-sight velocity weighted by the backscatter along the beam path within the seeded flow, rather than at a single point. This removes the need to consider range in the constrained tunnel environment. In a future iteration, focusing optics could be used to select a mea-

surement range, which would also concentrate the receiver sensitivity within the probe volume and improve the SNR.

Parameter	**Value**
Design wavelength	1550 nm
Focal length	37.13 mm
Numerical aperture	0.24
Output beam diameter ($1/e^2$)	≈7.0 mm
Full-angle divergence	≈0.28 mrad (0.016°)
Connector	FC/APC

Table 3.7: Key specifications of the Thorlabs F810APC-1550 fibre collimator. Beam diameter and divergence are theoretical values for SMF-28 fibre

3.5 Data Acquisition and Processing

As discussed in the introduction to this section, this project is not concerned with real-time processing of the data, which typically involves the use of high sampling-rate ADCs, and FPGAs which are programmed to be able to perform FFTs of the sampled data very quickly. Instead, data is obtained via a Keysight EDU1002X oscilloscope and post-processed on a laptop. The sampling rate of the oscilloscope is set to 125MHz, giving a Nyquist frequency of 62.5MHz. A beat signal at 62.5MHz would correspond to a speed of approximately 80m/s, which is far above any speed that will be measured, thus making the oscilloscope sampling rate more than sufficient.

Data is transferred from the oscilloscope to the laptop in discrete "records". Each oscilloscope record obtained is 25 000 samples long, corresponding to 200μs of data, however the oscilloscope can only obtain and transfer at a rate of approximately 8 records/second, while approximately 5000 records worth of data points are available for the same time period, meaning that only 0.2% of data for the data acquisition period is effectively used. By contrast, systems utilising FPGAs and high sampling-rate ADCs can sample up to 100% of the data in a given time period [13]. Thus, when considering the acquisition time for a measurement, the oscilloscope record obtaining bottleneck is not considered, and only the total amount of time which the data represents is considered. This is of importance, because in a real-world setup this time period determines how many wind-speed measurements the system can make over a given time-period, and thus determines how well it can detect rapid changes in wind-speed.

Using Python scripts, the power spectral density (PSD) of each capture is obtained with

Welch's method. Each record is divided into segments of N_{seg} samples that overlap by 50 %, and each segment is multiplied by a Hann window to suppress spectral leakage from its edges. The FFT of each windowed segment is then taken, and its squared magnitude gives a periodogram. Finally, the periodograms from all segments of all records are averaged to give the PSD. Averaging after the FFT in this manner reduces the variance of the noise floor, so that a weak Doppler peak can be distinguished from random fluctuations, with the tradeoff being that more data has to be acquired, over a longer time period.

The segment length sets the frequency resolution, $\Delta f = f_s/N_{\text{seg}}$, and, for a fixed record length, the number of segments, and hence the number of averages that are taken. Thus, for a fixed record length and number of records, there is a tradeoff between frequency resolution and number of averages (which help to reduce variance). For experiments A, B and C, $N_{\text{seg}} = 12\,500$ samples and 100 records of 25 000 samples are used, giving $\Delta f = 10\,\text{kHz}$ at $f_s = 125\,\text{MSa/s}$ (a velocity resolution of 7.75 mm/s). Each record then yields 3 overlapping segments, so a capture averages 300 periodograms. For the final wind-tunnel tests in section 5.2, where the weak aerosol return calls for more averaging rather than finer resolution, $N_{\text{seg}} = 2500$ and 500 records are used instead. This gives $\Delta f = 50\,\text{kHz}$ (3.9 cm/s), 19 segments per record, and 9500 periodograms in total.

Because the 50 % overlap makes adjacent Hann-windowed segments partially correlated, these correspond to about 245 and 7200 independent averages respectively. For stationary Gaussian noise, the standard deviation of each averaged PSD bin is $\sigma \approx S/\sqrt{N_{\text{eff}}}$, where S is the noise PSD and N_{eff} is the number of independent averages. This equation gives about $0.064\,S$ for the default settings and $0.012\,S$ for the wind-tunnel settings, illustrating that in the wind-tunnel case the noise variance will be approximately 5× lower. The absolute fluctuation of the noise therefore scales with the noise floor, while its relative size is set by the number of averages.

3.6 Final Design

The final design of the system using the architecture and components discussed above is given in Figure 3.2 below.

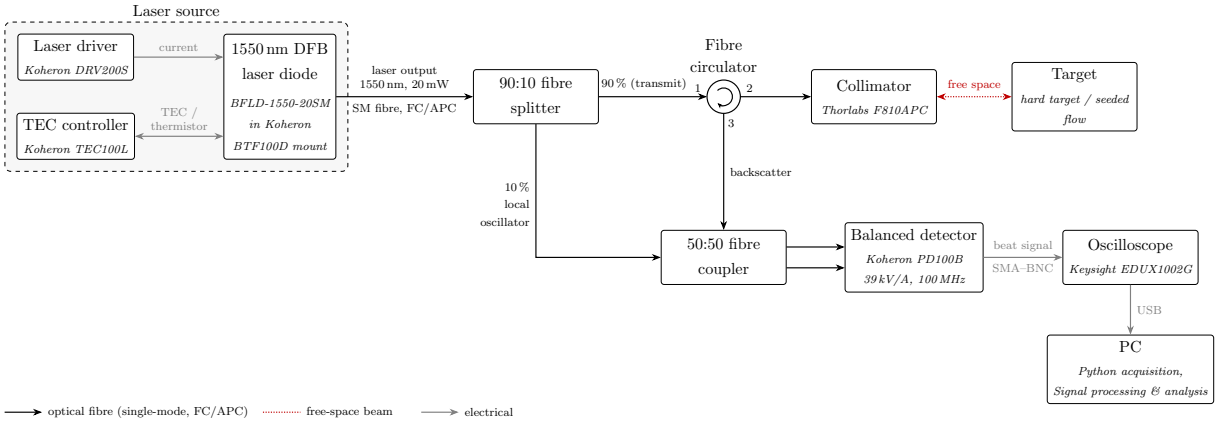


Figure 3.2: Architecture diagram of final system design

Chapter 4

Noise Analysis

4.1 Introduction

This chapter gives a comprehensive account of the testing which was done on the completed DWL system to account for the various sources of noise. The purpose of the noise analysis is to quantify the noise observed in the system and attribute it to the appropriate sources, while investigating how component, architecture, and operating point choice affects this. This chapter is separate from the functional testing of **??**, as the experimental setup differs significantly, and the results from these experiments will indeed be used to inform the expected results of the experiments in **??**, and to construct an optimal experimental setup in those cases.

4.2 A: Electronic and Shot Noise

4.2.1 Experimental Goals

As discussed in subsection 2.3.5, a coherent DWL performs best when it is shot-noise limited: the shot noise of the local oscillator (LO) must dominate every other noise source at the detector. Experiment A characterises the receiver's noise floor with the LO as the only light on the detector, so that the individual noise contributions can be separated and measured. It has the following goals:

- **Measure the electronic noise floor (A1).** With the laser off, measure the noise of the PD100B and acquisition chain alone, and compare it with the input-referred

noise specified in the PD100B datasheet. Agreement with the datasheet validates the data collection and post-processing chain used in all later experiments.

- **Measure shot noise as a function of LO power (A2).** Step the laser drive current to vary the LO power reaching the balanced detector, and confirm that the excess noise above the electronic floor grows linearly with LO power, as expected for shot noise.
- **Choose the operating point.** From the A2 results, find the LO power at which shot noise overtakes the electronic floor, and choose a laser operating point at which the system is close enough to the shot-noise limit to meet requirement P2 (Table 3.1).
- **Quantify RIN and validate balanced detection (A3).** Repeat the measurement with only one photodiode connected, making the PD100B a single-ended detector. Comparing this with the balanced case measures the laser's relative intensity noise (RIN) and confirms that the balanced detector cancels this common-mode noise as intended. This experiment is only included for completeness, to demonstrate RIN cancellation, and the results are given in Appendix A.2.

4.2.2 Experimental Methodology

The experimental setup used is given in Figure 4.1:

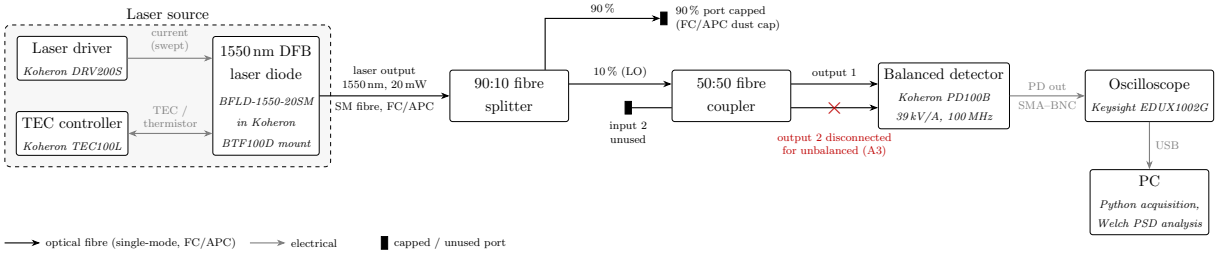


Figure 4.1: Experimental setup A

It can be seen that this differs from the full system setup. The circulator and collimator are unused for this experiment. The 90% splitter arm is left unconnected and is capped. Only the 10% splitter arm (LO) is connected to the one arm of the 50:50 coupler. Both outputs of the coupler are connected to the balanced detector, except in the case of experiment A3. Since there is no circulator, collimator or target, the only light reaching the detector is the LO itself. With no second optical field there is no beat term (as in Equation 2.5), and the LO contributes only a DC photocurrent, which is removed by the detector's AC coupling. Thus, the signal at the detector will be a pure noise signal, containing only electronic noise and shot noise. Note that phase-induced intensity noise is

not present here, as it originates from reflections in the collimator and circulator which are not present in this setup, and RIN is assumed to be cancelled fully by balanced detection.

The experiment is divided into three phases, with slightly different configurations for each in order to give different measurands, as shown in Table 4.1:

ID	**Configuration**	**Quantity Measured**
A1	Laser output off, detector switched on	Electronic noise only
A2	Laser output on (current varied), detector switched on, both detector inputs connected	Electronic noise + shot noise
A3 (supplementary experiment)	Laser output on (current varied), detector switched on, only one detector input connected	Electronic noise + shot noise + full RIN

Table 4.1: Experiment A phases and configurations

For each measurement the laser is held at a fixed drive current and 100 records of the detector output are captured sequentially with the oscilloscope and transferred to a laptop. The averaged PSD of the record is estimated, as described in section 3.5. A1 requires only a single measurement, while A2 and A3 are repeated over a range of drive currents up to the 48 mA operating point.

To compare the results with a single figure, the PSD can be averaged over the 5–20 MHz band. From Equation 2.1, this band corresponds to target speeds of roughly 4–15.5 m/s, making it a reasonable band to take an average noise power measurement over, as it is where the Doppler peaks in subsequent tests would appear.

4.2.3 Results and Discussion

A1: Electronic Noise

The PSD for the measurement was obtained which gives the electronic noise from the detector only. The average PSD over the 5 - 20MHz band is calculated (using the script) as:

$$PSD_{avg(5-20MHz)} = 8.06 \times 10^{-14} V^2/Hz \tag{4.1}$$

Using the PD100B detector's gain value of 39kV/A, this can be converted into an average input noise current density across 5 - 20MHz, for comparison to the PD100B datasheet:

$$I_{input} = \frac{8.06 \times 10^{-14} V^2/Hz}{(39kV/A)^2} = 7.28pA/\sqrt{Hz} \tag{4.2}$$

Input noise current density can be seen as the equivalent of injecting this noise current at the detector input. The calculated average of $7.28pA/\sqrt{Hz}$ across 5 - 20MHz agrees closely with the PD100B's datasheet value of $8pA/\sqrt{Hz}$, measured at 10MHz [**?**]. Thus, this level of electronic noise is expected, and the measurement and post processing chain is validated.

In order to achieve shot-noise limited operation, the average PSD of the shot-noise over this region must thus be $\gg 8.06 \times 10^{-14} V^2/Hz$.

A2: Electronic Noise & Shot Noise

Figure 4.2 gives the PSD plots of the total noise - made up of shot-noise, electronic noise, and residual RIN - for different laser currents on the same axes:

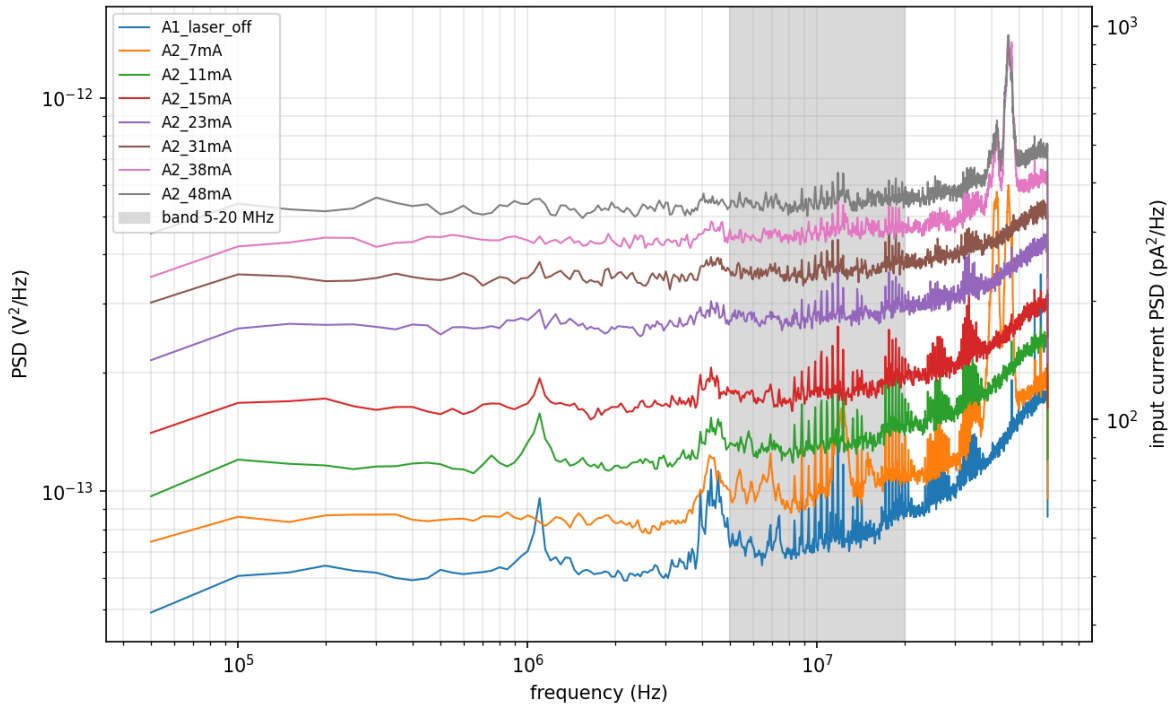


Figure 4.2: PSD of total noise (shot, electronic and relative intensity noise) for different laser source currents

It can be seen that the total noise increases with increased laser current - this is expected, due to the linear relationship between shot noise and LO power (which is related to laser current which is varied here). In order to compare the PSDs for the various laser

currents, the average PSD across the 5 - 20MHz range can again be taken here, for each laser current. Figure 4.3 gives these averages plotted against laser current:

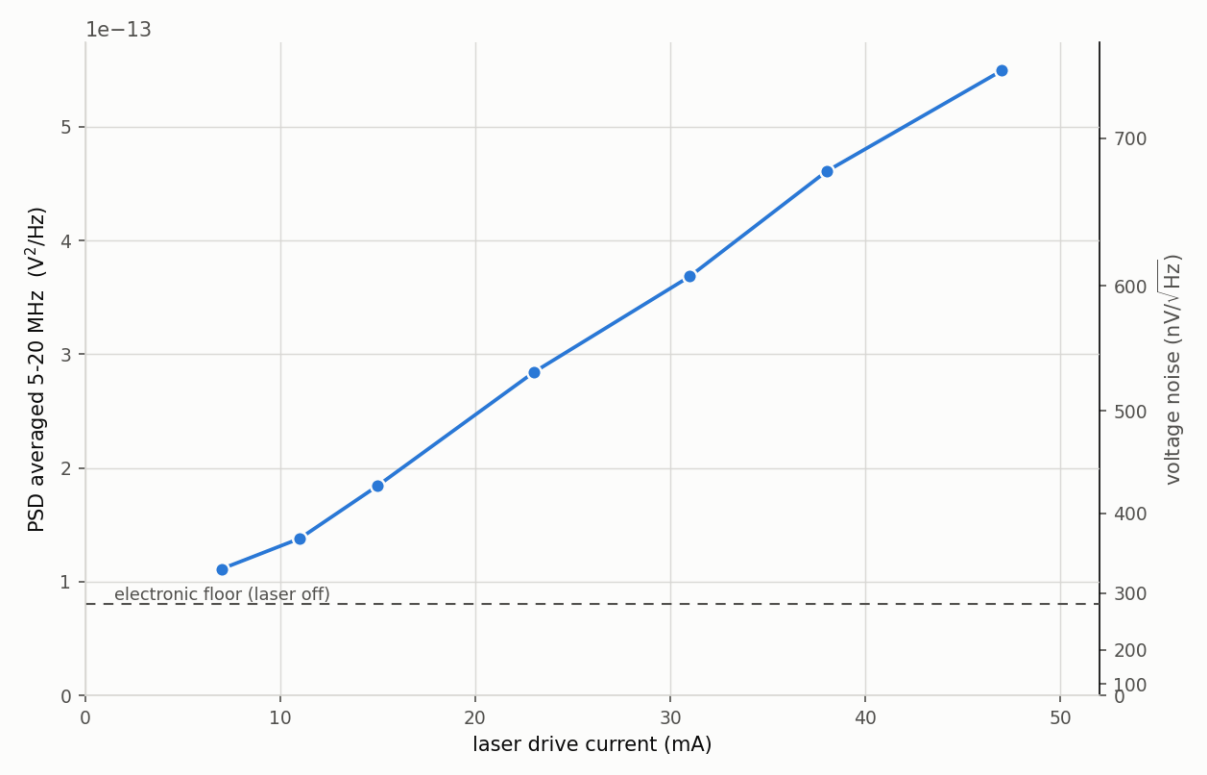


Figure 4.3: 5 - 20MHz Average PSD vs Laser Current with Balanced Detection

It can clearly be seen in Figure 4.3 and Figure 4.2 that the shot-noise exceeds the electronic noise floor at even the lowest measured laser currents, thus making the system shot-noise limited. As the current is increased, the system approaches the ideal shot-noise limit, as shot noise increasingly swamps out the electronic noise floor.

Performance requirement P2 in Table 3.1 requires that the total detector noise is within 1dB of the shot-noise limit. This requirement can be decomposed as follows:

Noise power adds in linear units, so letting S be shot noise, and E be every other noise source present here (electronic noise and any remaining RIN), the total noise floor, T is given by:

$$T = S + E = S(1 + \frac{E}{S}) \tag{4.3}$$

Converting to dB:

$$10log_{10}(S(1 + \frac{E}{S})) = 10log_{10}(S) + 10log_{10}(1 + \frac{E}{S}) \tag{4.4}$$

In this form, it can clearly be seen that as S increases, the effect of E on the total

noise floor is increasingly reduced, as the second term, $10log_{10}(1 + \frac{E}{S})$ approaches zero. This second term is the excess noise above the ideal shot-noise limit, which is given by $10log_{10}(S)$. Thus, in order to obtain a total noise floor which is within 1dB of the shot noise limit, we set:

$$10log_{10}(1 + \frac{E}{S}) = 1dB \tag{4.5}$$

$$\rightarrow \frac{E}{S} \leq 0.26 \tag{4.6}$$

Substituting the obtained electronic noise floor obtained in A1 into Equation 4.6, we obtain:

$$S \geq 3.1 \times 10^{-13} V^2/Hz \tag{4.7}$$

Noting that Figure 4.2 and Figure 4.3 both give the **total** detector noise floor (shot **and** electronic noise), we require the total noise to be above $3.906 \times 10^{-13} V^2/Hz$ to gives a shot noise value of $3.1 \times 10^{-13} V^2/Hz$. This can be seen to be reached at a laser current of approximately 32mA. Above 32mA, the total detector noise floor is within 1dB of the ideal shot-noise limit, and the ideal shot-noise limit is further approached as the current is increased.

From this, it can be seen that an operating current of 48mA for the laser allows P2 in Table 3.1 to be met with the highest margin, and this will thus be used as the laser current for the remainder of this project. While a higher laser current still may be used, to even further approach the ideal shot-noise limit, this risks saturating the detector. The detector has a maximum incident power rating of 1.5mW on each photodiode [**?**]. From the laser datasheet [**?**], a current of 48mA gives approximately 17mW at the laser output, which will result in approximately 0.85mW at each detector from the LO alone. This allows the detector to thus remain comfortably far away from its saturation point, even if returned signal is high and results in a higher incident power at the photodiode after mixing with the LO.

A3: Relative Intensity Noise

Experiment A3 serves only for completeness, to illustrate the effectiveness of balanced detection in removing relative intensity noise at the detector output. It was found that indeed 0.6125dB of RIN was cancelled by using the balanced detection setup. See Ap-

pendix A.2 for the full results and discussion of this supplementary experiment.

4.3 B: Phase-induced Intensity Noise

4.3.1 Experimental Goals

As discussed in subsection 2.3.5, and elaborated in Appendix A.3, Phase-induced Intensity Noise (PIIN) can be a significant contributor to the total noise of the system, rising far above shot-noise - particularly when the linewidth of the laser source is not narrow, and when there is significant path-length mismatch between the LO and the transmit signal.

The goal of Experiment B is to measure and quantify the magnitude and shape of the total noise of the system, including PIIN, in a controlled setting. There is effectively no target in this experiment, thus making all PIIN in the measurement attributable to reflections from the circulator and collimator, and independent of target.

4.3.2 Experimental Methodology

For this experiment, the system is set up in full, as given by the final design in Figure 3.2. The collimator is then pointed at a stationary matte-black painted surface, angled 45 degrees away from the collimator, and at a distance of 10cm from the collimator. Any reflections from the target surface will not be Doppler shifted, because it is stationary - thus, if these reflections are sent back into the collimator, they will appear at the output as additional PIIN (see Appendix A.3). This setup thus intends to absorb and reflect a large amount of the emitted light **away** from the system, so that the measured PIIN can be attributed to only the internal reflections in the system from the collimator and circulator, which is independent of target.

The laser is then driven with a current of 48mA (the chosen operating point for LO power), and the same oscilloscope acquisition chain and post-processing chain (Welch PSD) as in experiment A is used to obtain the PIIN power. As discussed in [15], PIIN can vary drastically depending on temperature, or small disturbances in the system's fiber optics, thus this measurement is repeated 10 times over the course of an hour to account for such variations. The average noise power as well as the spread from these measurements can then be computed.

4.3.3 Results and Discussion

Initial Results

The total noise was found to be significantly above that measured in A2, owing to the excess noise introduced by PIIN. In the 5–20 MHz band, the total noise was measured to be 22.1 dB above the shot-noise floor. PIIN is thus the largest single noise contributor in the system, and the system is therefore no longer shot-noise limited. Experiments C and D (Chapter **??**) were first attempted using the system as is, with the PIIN-limited behaviour described above. While the system could adequately detect the velocity of a hard target, it did not have enough sensitivity to detect the velocity of a seeded air-flow, owing to the high noise floor from PIIN. It was thus decided that an attempt would be made to reduce PIIN, from the theory discussed in subsection 2.3.5 and elaborated in Appendix A.3, and the experiment would be rerun for this improved system.

Reducing PIIN: Methodology and Results

Obtaining a narrower linewidth laser source is out of budget for this project, so the only remaining option was to better match the fibre optic path length between the LO and the received signal. This allows the LO and the received signal to be more coherent in phase at the mixer, as the phase of the laser will not have been able to "wander" by as much between the two signals. FC/APC fibre patch cords of varying lengths were inserted between the 10% splitter arm and the 50:50 mixer, to increase the LO path length to more closely match the longer path followed by the received signal (which traverses the collimator and free space twice, in excess of the LO path). The total system noise was measured for each patch cable length using the same methodology, with 10 captures each. The results are given in Figure **??** and Table **??**, with the electronic noise floor and the total noise floor (excluding PIIN) from Experiment A2 given as reference.

Configuration	**Mean PSD** ($10^{-12}\,\mathrm{V}^2/\mathrm{Hz}$)	**Above shot noise (dB)**	**Measured reduction (dB)**	**Predicted reduction (dB)**
No matching	85.5	22.1	0.0	0.0
2 m cable	27.9	17.2	−4.9	−5.6
3 m cable	18.4	15.4	−6.8	−10.9
4 m cable	1.98	5.7	−17.7	−26.4

Table 4.2: Total system noise in the 5 MHz to 20 MHz band for each LO-arm patch cable. The measured reduction is that of the PIIN alone (total minus the balanced shot-noise reference), and the predicted reduction is $20\log_{10}(\Delta L/\Delta L_0)$, from the ΔL^2 dependence of Harris et al. [11], where ΔL is the path mismatch and $\Delta L_0 = 4.2\,\mathrm{m}$ is the initial mismatch estimated from the circulator fiber length

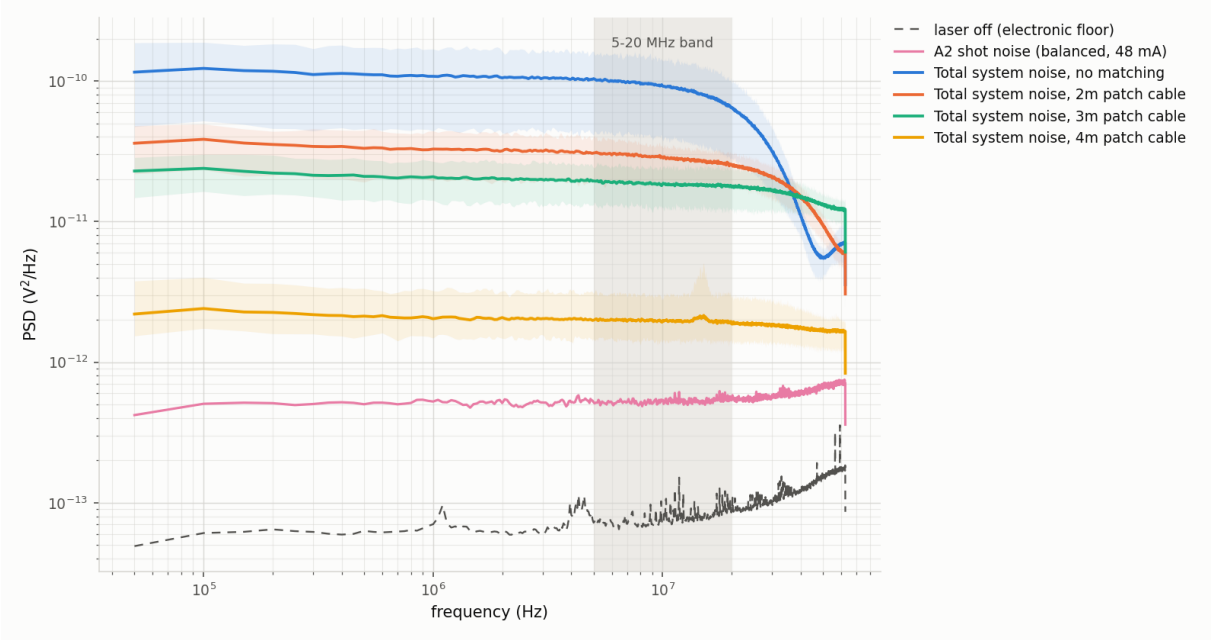


Figure 4.4: Total noise in the system for different inserted patch-cable lengths

As expected from the theory of Harris et al. and Jiang et al. [11] [15], PIIN varied significantly, even between measurements with the same setup. It can also be observed that the shape of the PIIN corresponds to the $L_c \gg \Delta L$ regime described in Appendix **??**, as it is a flat floor and does not contain a Lorentzian peak at baseband.

It was found that increasing the patch cable length (and thus increasing the fibre path matching) significantly reduced the total noise, with the PIIN reduction loosely matching the ΔL^2 dependence identified by Harris et al. [**?**]. The 4m patch cable gave a remarkable 17.7dB reduction in PIIN, bringing the total noise to just 5.7dB above shot noise. While the system is still PIIN-limited, it is far closer to the ideal shot-noise limit, and this gain in sensitivity is what made the wind-speed measurements of Experiment D possible. The

4 patch cable is therefore kept in the system for all results reported in chapter 5.

Chapter 5

Functional Testing

This chapter details the testing which was done on the system's primary function: velocity measurement. Two distinct test setups are used, which are described in this section - results were then collected for these test setups and are presented and discussed with respect to the goals of the system and the overarching project.

5.1 C: Hard-target Speed Measurement

5.1.1 Experimental Goals

Before attempting wind-speed measurement, where the aerosol return signal is very weak, the working principle of the system is first to be validated on a hard target, whose return signal is far stronger and whose velocity can be set precisely. This isolates set-up and processing problems from the difficulties of a weak signal, and tests performance requirements P1 (hard-target SNR), P5 (low-speed detection) and P7 (velocity accuracy).

5.1.2 Experimental Methodology

A disc driven by a NEMA-17 stepper motor provides a constant, accurately known target speed. The 3D-printed rig in Figure 5.1 holds the collimator at $\alpha = 45°$ to the motor axis, aimed at a point a distance $d = 15\,\text{mm}$ from the disc centre. Since the disc is at an angle to the collimator, there will be a component of its angular velocity which is in the

direction (line-of-sight) of the beam. The line-of-sight velocity at that point is

$$v_{\mathrm{LOS}} = \frac{2\pi N}{60}\, d \sin\alpha, \tag{5.1}$$

where N is the motor speed in RPM. Two features of the geometry affect the measurement. Any error in d or α scales v_{LOS} directly. Additionally, the collimator output beam is **not** infinitesimally small, and thus the finite beam spot (which is about 7 mm in diameter [**?**]) covers a range of radii on the disc, and hence covers a number of line-of-sight velocities, which broadens the Doppler peak.

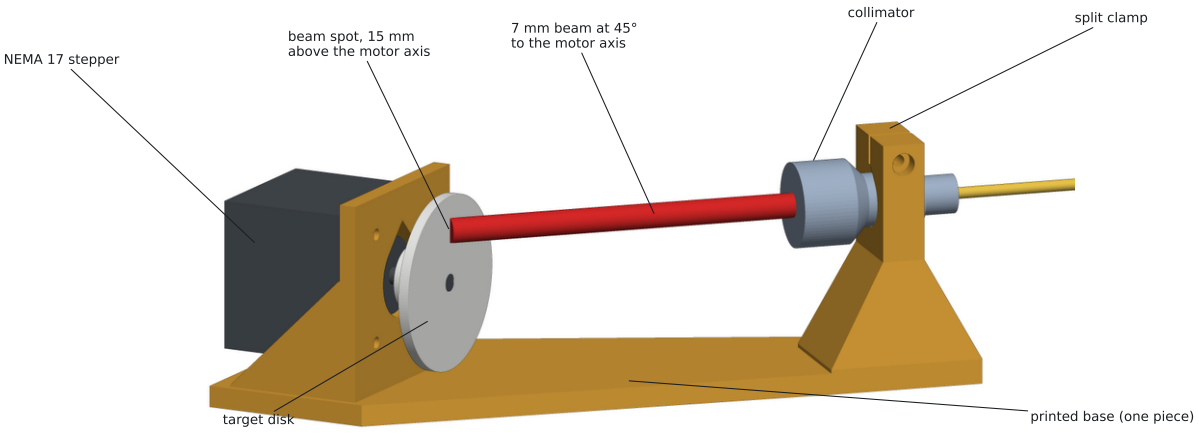


Figure 5.1: Test setup used for hard-target speed measurement

The PLA disc transmits much of the light at 1550 nm, allowing returns from stationary parts behind it, so white paper was attached to its face to provide a strong, diffuse return. The rest of the system was configured as in Figure 3.2, with the laser at 48 mA. For each motor speed, 100 oscilloscope records were captured, and their Welch PSDs averaged, using the same processing chain as experiments A and B, as detailed in **??**sec:design-DataAcquisition). From Equation 2.2, the expected Doppler shift is 1.29 MHz per m/s of v_{LOS}.

5.1.3 Results and Discussion

Figure 5.2 below shows the PSD of the measurements obtained for different motor RPMs tested. Table **??** gives the Doppler peak and local noise floor for each target speed. The Doppler peak, f_D, is taken as the power-weighted centroid of the excess PSD at each peak, as compared to the stationary-target reference, and $v_{\mathrm{LOS}} = f_D/(1.29\,\mathrm{MHz\,s/m})$. The noise statistics are taken in a band outside 3×FWHM around the centroid, out to 2 MHz either side. All PSD values are in the same units, so the PSD at f_D can be

compared directly with the detection threshold (mean $+5\sigma$). The excess power created at the Doppler frequency is calculated by integrating each peak across the frequencies which it spans.

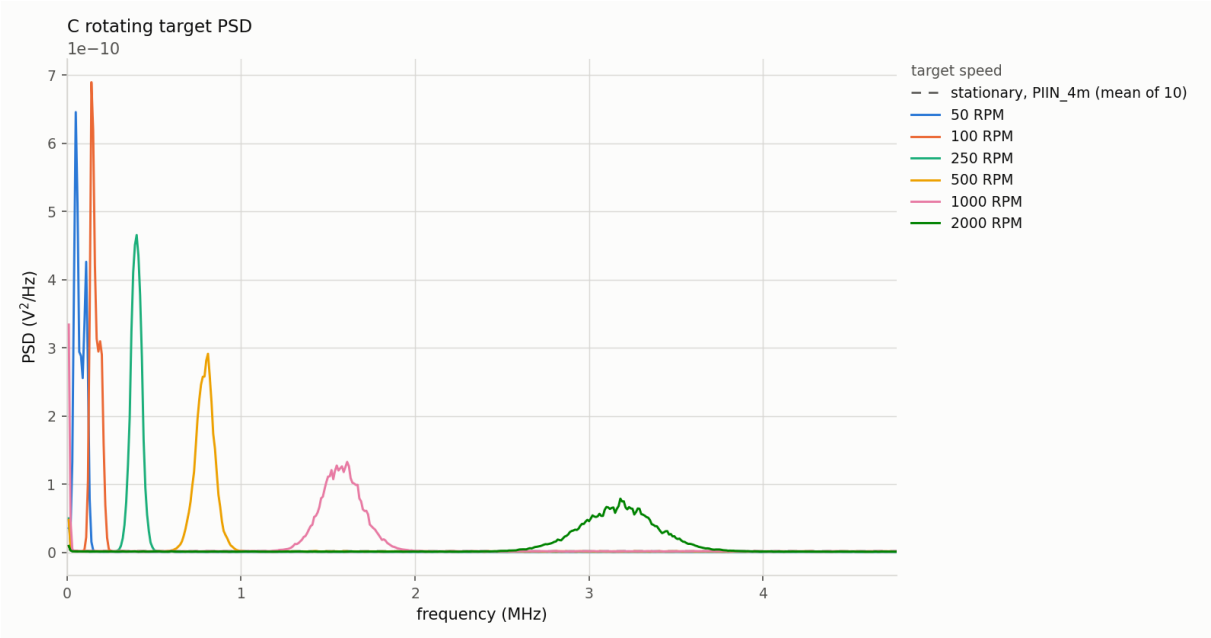


Figure 5.2: C: PSD of hard-target speed measurements, at different motor RPMs

Speed (RPM)	f_D (MHz)	v_{LOS} (m/s)		**PSD** (10^{-12} V^2/Hz)			**Peak PSD/ mean** (dB)	**Excess power** (10^{-5} V^2)
		Measured	Expected	Mean	Mean $+5\sigma$	**At f_D**		
50	0.074	0.057	0.056	1.15†	1.57	294.1	24.1	3.65
100	0.162	0.126	0.111	1.38†	1.98	421.3	24.9	3.65
250	0.397	0.308	0.278	1.20†	2.09	465.3	25.9	3.57
500	0.793	0.615	0.555	1.68†	3.99	257.9	21.9	3.58
1000	1.577	1.222	1.111	1.84	3.03	126.2	18.4	3.61
2000	3.174	2.460	2.221	1.39	2.54	68.0	16.9	3.50

Table 5.1: C: PSD Doppler peak and noise statistics for each motor RPM tested

Noise statistics exclude bins below 50 kHz (near-DC residual from disc imperfections).

† Peak too close to DC for a full noise band below it; the floor is taken mostly or entirely from the high-frequency side and is likely a slight underestimate.

A distinct Doppler peak was obtained at every speed, between 16.9 dB and 25.9 dB above the local noise floor and well clear of the mean $+5\sigma$ detection threshold in every case, satisfying P1. The peak height falls and its width grows with speed because the velocity spread across the beam spot scales with N. However, the integrated excess power stays constant at $(3.6 \pm 0.1) \times 10^{-5}$ V^2, which shows that the returned power is unchanged and is merely spread over a wider band. At 50 RPM ($v_{LOS} = 0.056$ m/s) the peak is still

clearly resolved, satisfying P5; here however, the 10 kHz bin width of the Welch PSD (corresponding to 7.75 mm/s of velocity) and the proximity to DC limit the precision, which explains why this point does not show the systematic error seen at higher speeds, as discussed below.

It can also be seen that f_D, the centroid of each Doppler peak, and its corresponding measured velocity, $v_{LOS} = \frac{f_D}{1.29MHz}$, was higher than the predicted Doppler peak, by approximately 10% for each case, causing the system to fail performance requirement P7 for the hard-target case. This illustrate visually in Figure 5.3:

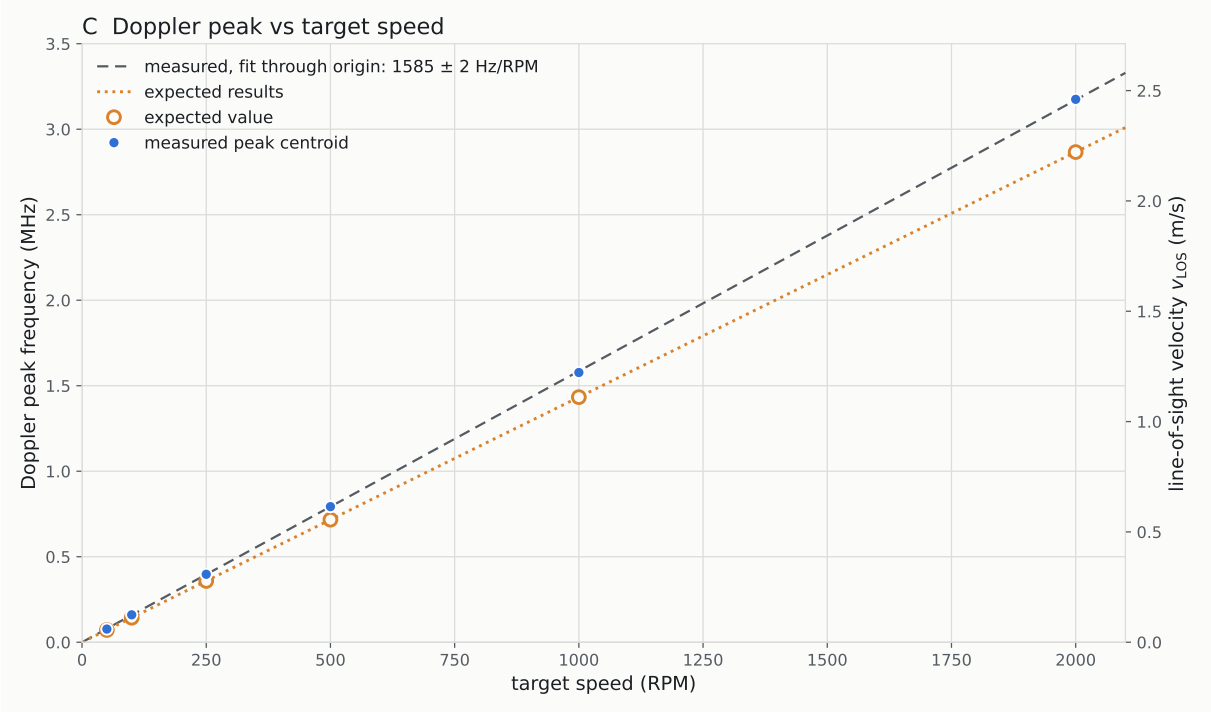


Figure 5.3: C: Measured Doppler peak frequency (from centroid) and expected Doppler peak frequency vs target motor RPM

As Figure 5.3 shows, f_D is linear in N: a fit through the origin gives 1585(2) Hz/RPM, and all points from 250 RPM upwards lie within 0.5 % of it. However, the slope is 10.6 % above the expected 1433 Hz/RPM, so P7 is not met as built. Because the error is a constant scale factor, it points to the geometry rather than to the measurement itself. Since v_{LOS} is proportional to d, the beam striking the disc only 1.6 mm further from the axis than intended would account for it. Over the 100 mm collimator-to-disc distance, that is a collimator tilt of only about 0.9°. Adjusting the collimator bracket screw changed the percentage error between experiment sets, while it remained constant within each set, which supports this explanation.

In summary, the experiment validates the Doppler measurement principle: the system resolves speeds from 0.06 m/s to 2.5 m/s, with a linear response and a large SNR margin

on a hard target. The 10.6 % offset is a calibration error in the rig geometry, which a single-point calibration would remove. The aerosol return in experiment D is expected to be far weaker, so this margin does not by itself guarantee wind-speed measurement, which is addressed in section 5.2.

5.2 D: Wind-speed Measurement

5.2.1 Experimental Goals

Now that the noise analysis of the system is complete, and the speed-measurement principle has been shown to be functional for a hard-target, the system can finally be tested for seeded air wind-speed measurement. The primary subject of investigation for this section is the sensitivity of the system with respect to its noise and the different levels of seeding in the air, as well as its accuracy at different tested speeds. As discussed previously, the range of the system can not be tested in a wind-tunnel environment, and thus this is out of scope for this project.

5.2.2 Experimental Methodology

In order to test wind-speed measurements in a controlled environment, a test-setup was built in an existing wind-tunnel. The wind-tunnel is capable of producing measured, stable flows at 3.5 - 20m/s, with the flow-speed being measured at the centre of the test section. The tunnel has an inlet, where an air seeding device can be placed to seed the tunnel air. Two types of seeding are investigated; "fog" machines, which use mixtures of glycol or glycerin, and distilled water to produce a dense, visible vapour cloud in the air (of the same type typically seen in entertainment venues), and "misters", which simply use water to generate a very fine (barely visible) mist in the air, which is akin to atmospheric conditions on a misty day. It must be noted that at higher tunnel wind-speeds, the atmospheric seeding becomes more diffuse, and it is thus expected that for the same amount of seeding the Doppler return will decrease at higher speeds.

The test setup used in the wind-tunnel is shown in Figure **??**:

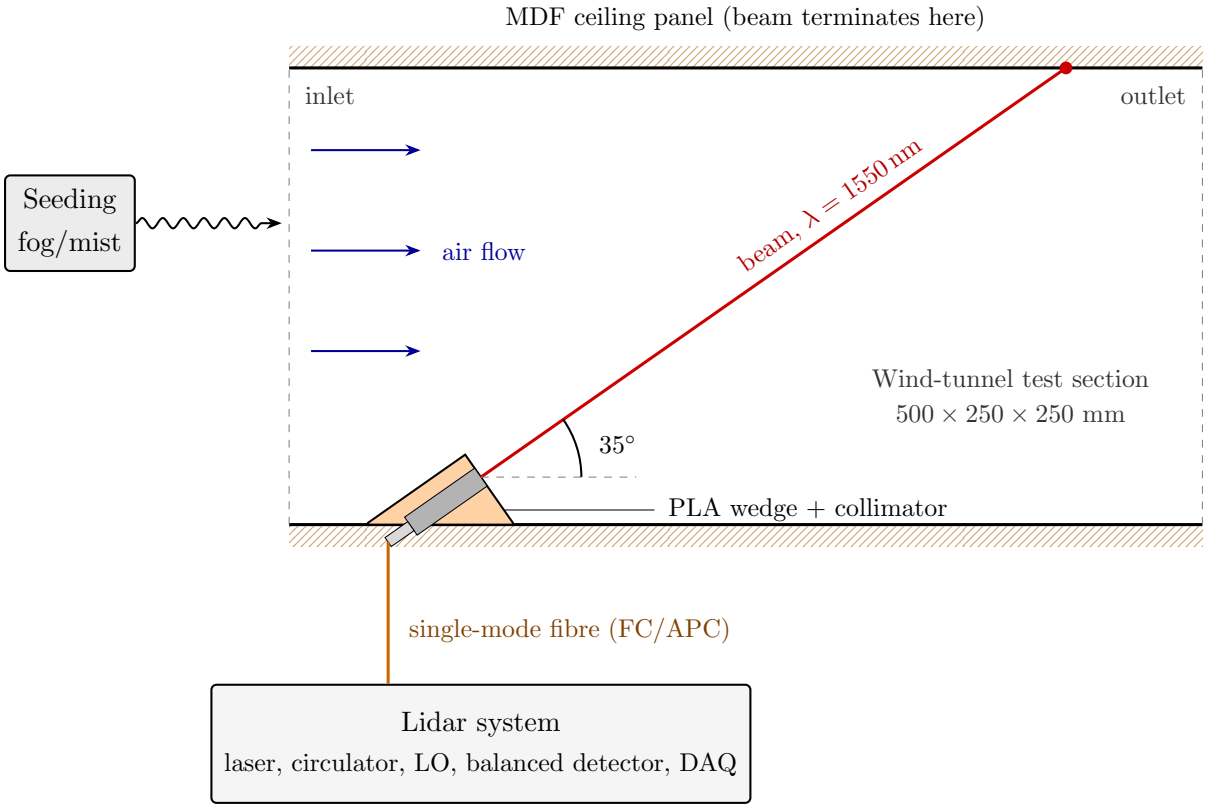


Figure 5.4: D: Wind-tunnel experimental setup diagram

A small wedge which fits the collimator and which can be attached to the floor of the wind-tunnel was designed and 3D printed. The collimator is pointed in the direction of the flow, so as to minimise any residue buildup from seeded air hitting the collimator lens from front-on. The consequence of this however, is that the collimator has to be placed at an angle (35 degrees in this design), rather than straight in the direction of flow, so as to avoid the collimator and the housing from disturbing the flow and causing excessive turbulence, immediately prior to the measurement line-of-sight. This would indeed not as heavily be the case if the collimator was pointed **opposing** the flow, as measurements would then be taken in a line-of-sight **before** the housing. The angled wedge however also provides the advantage of allowing the beam to terminate on a stationary angled surface (the MDF ceiling of the wind-tunnel), rather than continuing on through the tunnel to hit the rotor which generates the flow, which would generate additional Doppler components at the radial velocity of the rotor-blades in the direction of the beam. The angled beam termination surface also serves to absorb and reflect unwanted stationary (non-Doppler shifted) beam components away from the system, so that there is minimal additional PIIN - this mimics the setup of section 4.3.

It must be noted from the rig geometry, that the system only measures the component of the velocity which is in the direction of the beam. Assuming no vertical flow in the

tunnel, the velocity "seen" by the beam is:

$$v_{los} = cos(35) \times U \tag{5.2}$$

Where U is the horizontal flow-speed in the wind-tunnel, as measured by its sensor. As previously derived, a 1550nm beam will be Doppler shifted by approximately $1.29MHz$ per m/s of flow in the direction of the beam. Thus, for this test setup, a Doppler shift of

$$\Delta F = 1.29MHz \times cos(35) \times U = 1.057MHz \times U \tag{5.3}$$

is expected, i.e. $1.057MHz$ per m/s of wind-tunnel flow.

Again, the system is fully set up for these experiments as for C, as shown in Figure 3.2 and with a laser drive current of 48mA. The Welch method for calculating PSD was again used here, as for experiments A, B, and C, however in this case 500 oscilloscope records are obtained, and each segment length is set to 2500, giving a total of approximately 7200 independent averages (see section 3.5). This gives a lower noise variance which will make the detection of a smaller Doppler peak easier, in the absence of a laser power amplifier which would increase the aerosol return, and thus increase the size of the Doppler peak.

Even with this high amount of averages, the 500 records of 25 000 samples at 125 MSa/s represent only 100 ms of sampled data, which is comparable to the 4 ms to 41 ms integration times of established CW lidars[13][4]. Each 200 µs record is short enough that the flow is effectively frozen within it, so with a continuously streaming digitiser, with an FPGA (or similar) to process the data in real-time, the same averaging would give a new velocity estimate every 0.1 s. This 10 Hz update rate is well above the 0.5 Hz to 1 Hz found to be sufficient for lidar-assisted wind turbine control [18] [23], though below the 50 Hz of the commercial ZephIR [13]. In this set-up, however, the oscilloscope's transfer bottleneck spreads the records over roughly 35 s, so each PSD is an average of the flow over that period. This is acceptable here because the tunnel speed is held constant during each capture, but in a field-deployed iteration of the system an FPGA based acquisition chain should certainly be investigated.

5.2.3 Results and Discussion

Figure 5.5 below shows the PSD of the data obtained for different wind-tunnel setpoints, with dense fog-seeded air. It can be seen that there is a large variation in the noise floor between the experiments, making it difficult to clearly distinguish the different curves, thus a second plot is provided alongside the raw PSD, where each PSD is normalised by its respective noise floor, so as to make the results more clear. See section A.4 for the raw

PSDs of each case plotted on their own axes. Table **??** gives the Doppler peak and local noise floor for each wind-tunnel setpoint, using the same method of computation for the noise floor as was used in the hard-target case in section 5.1.

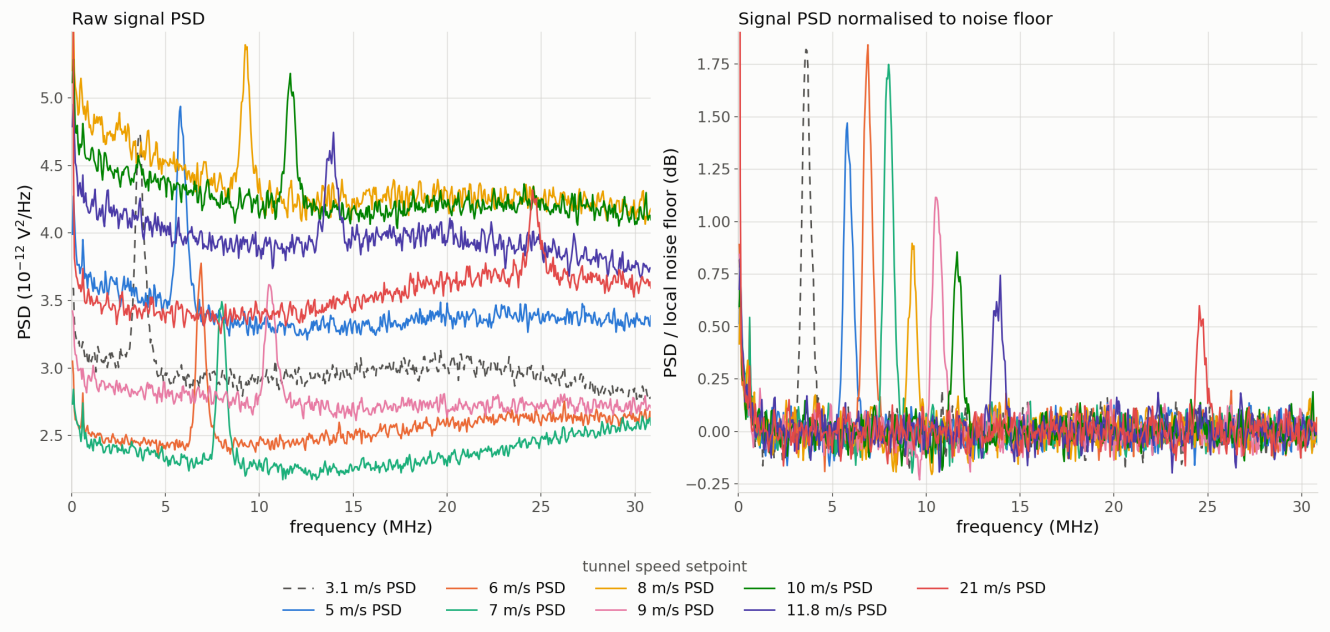


Figure 5.5: D: PSD for wind-speed measurements, at different wind-tunnel setpoints, with dense fog-seeded air

Setpoint (m/s)	f_D (MHz)	v_{LOS} (m/s)		**PSD** (10^{-12} V²/Hz)			**Peak PSD/ mean** (dB)	**Excess power** (10^{-7} V²)
		Measured	Expected	Mean	Mean $+5\sigma$	**At f_D**		
3.1	3.637	2.819	2.539	3.01	3.41	4.73	1.96	10.25
5.0	5.868	4.548	4.096	3.45	3.82	4.79	1.43	8.87
6.0	6.892	5.341	4.915	2.44	2.59	3.78	1.90	7.55
7.0	7.997	6.198	5.734	2.30	2.50	3.49	1.82	7.04
8.0	9.280	7.192	6.553	4.30	4.71	5.36	0.95	5.99
9.0	10.611	8.224	7.372	2.74	2.95	3.60	1.19	6.08
10.0	11.664	9.040	8.192	4.20	4.46	5.18	0.92	6.00
11.8	13.837	10.723	9.666	3.94	4.20	4.51	0.59	5.27
21.0	24.674	19.122	17.202	3.69	3.95	4.22	0.59	4.01

Table 5.2: D: PSD Doppler peak and noise statistics for each wind-tunnel setpoint, with dense fog-seeded air

Expected $v_{LOS} = U\cos 35°$; measured $v_{LOS} = f_D\lambda/2$. Noise statistics are taken over a 3×FWHM region either side of the peak out to 2 MHz, excluding bins below 50 kHz.

It can be seen that a distinct Doppler peak was obtained for every wind-tunnel setpoint, which has a power above the noise floor's mean $+5\sigma$ threshold, thus satisfying performance

requirement P6. The peaks were significantly lower than for the hard-target case, as expected, ranging from only 0.59 - 1.96dB above the mean of the noise. It can also be seen that the Doppler peak power decreases with increased wind-tunnel speed - this is because the increased wind-speed makes the seeding in the air more diffuse, resulting in lower returned power, and thus giving a smaller distinguishable peak. Indeed, integrating the excess power of each peak (shown in the last column of the table) supports this explanation, as the excess power decreases with increased wind-speed.

It can be seen that each peak corresponds to a line-of-sight velocity, v_{LOS}, which is consistently 8.1 - 11.6% higher than the expected line-of-sight velocity at that wind-tunnel speed setpoint. The wind-tunnel's wind-speed sensor was compared against an accurate pitot-tube anemometer in order to verify that the tunnel's setpoints are accurate, and it was found that this was the case, thus ruling setpoint error out as a possible cause of the DWL system's approximate 10% overshoot in all measurements.

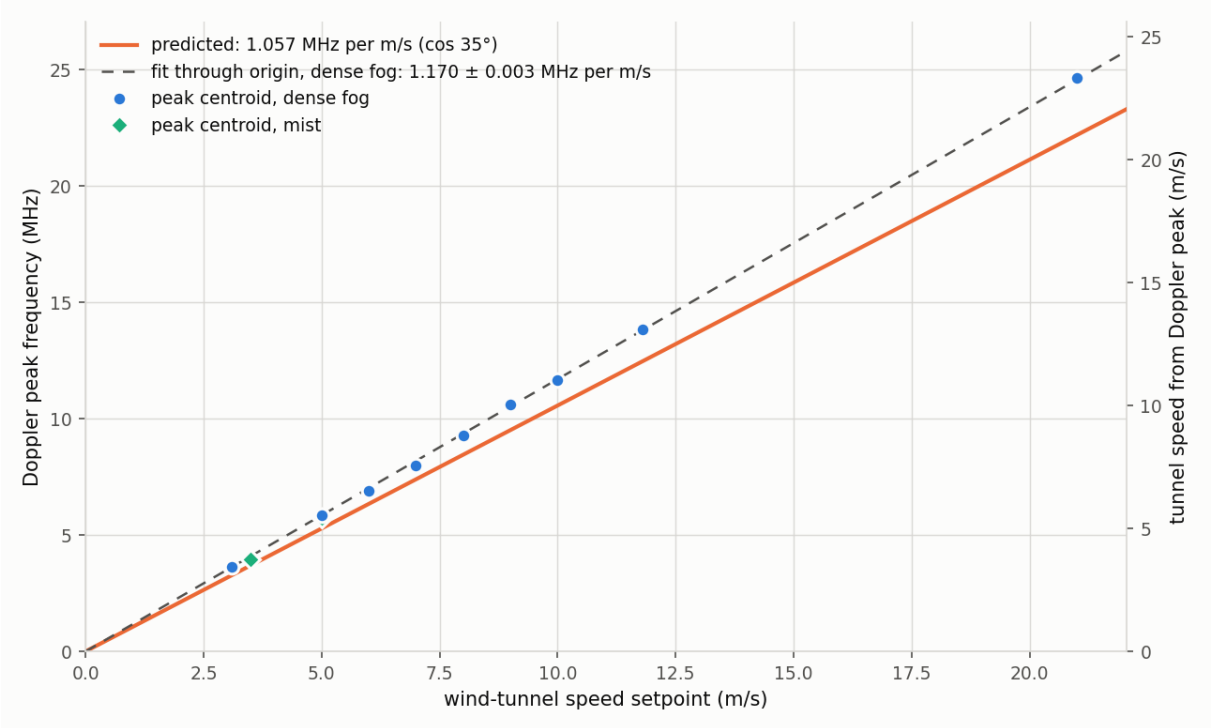


Figure 5.6: D: Measured Doppler peak frequency (from centroid) and expected Doppler peak frequency vs wind-tunnel speed setpoint

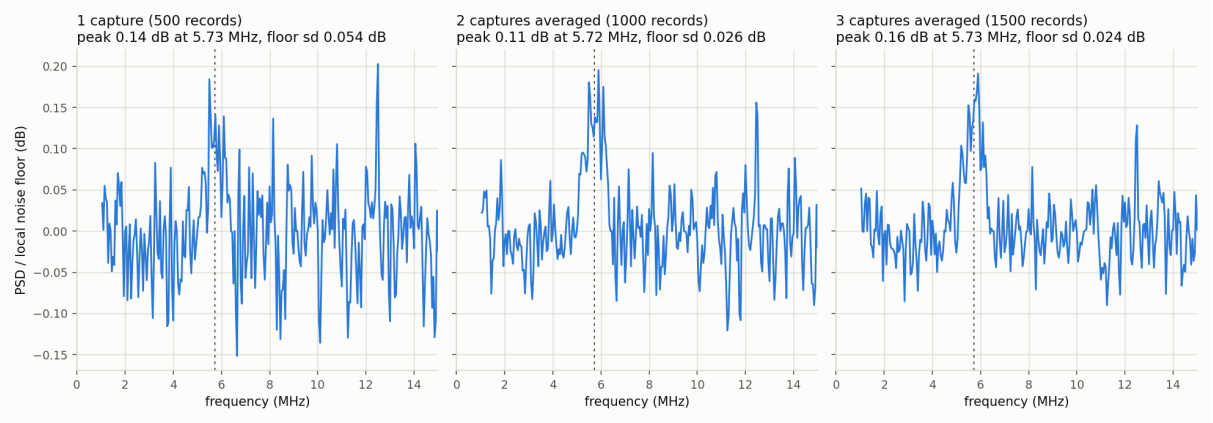


Figure 5.7: D: PSD for 5m/s wind-speed measurement, with light mist-seeded air. Subfigures show PSD for different numbers of averaged records to illustrate reduction in noise variance.

Table 5.3

Speed (m/s)	**Captures averaged**	f_D (MHz)	**FWHM** (MHz)	v_{LOS} (m/s)		**PSD/floor** (dB)			**Peak** (σ)
				Measured	Expected	At f_D	Mean	σ	
3.5	1	3.938	0.85	3.052	2.867	0.695	−0.050	0.0709	10.5
3.5	2	3.956	0.90	3.066	2.867	0.694	−0.042	0.0480	15.3
5.0	1	5.730	0.80	4.441	4.096	0.141	−0.017	0.0543	2.9‡
5.0	2	5.716	1.00	4.430	4.096	0.114	0.000	0.0262	4.4‡
5.0	3	5.726	0.75	4.438	4.096	0.160	−0.005	0.0243	6.8

PSD/floor is each PSD divided by its own running median (61 bins); captures are averaged in power. Expected $v_{\mathrm{LOS}} = U\cos 35°$; measured $v_{\mathrm{LOS}} = f_D\lambda/2$. Noise band as in Table **??**. Peak (σ) is (peak − mean)/σ.

‡ Below the 5σ detection threshold.

Chapter 6

Discussion

Here is what the results mean and how they tie to existing literature...

Discuss the relevance of your results and how they fit into the theoretical work you described in your literature review.

Chapter 7

Conclusions

These are the conclusions from the investivation and how the investigation changes things in this field or contributes to current knowledge...

Draw suitable and intelligent conclusions from your results and subsequent discussion.

Chapter 8

Recommendations

Make sensible recommendations for further work.

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Appendix A

Appendix A

A.1 Balanced Detection: RIN Cancellation

As discussed in subsection 2.3.5, the best-case performance of a DWL system is for it to operate in a shot-noise limited regime - thus, all other sources of noise should be reduced to below the shot-noise level. In coherent detection LiDAR architectures, the use of a balanced detector setup has become common for this purpose, as it assists in reducing a laser source's relative-intensity noise (RIN) below the shot-noise level, in a manner which single-ended detection does not.

A single-ended detection system simply uses a 2:1 (2 inputs, one output) fiber optic coupler to mix the LO reference signal with the backscatter signal, and uses a single photodetector to convert this light signal into a measurable current. It can be seen that this detection method retains the DC terms P_{LO}, P_{RS}. Any RIN from the laser source is present in the output, as these intensity fluctuations are contained in these two power terms. Thus, the measurable Doppler peak will be smaller with respect to the noise floor.

Contrastingly, balanced detection setup makes use of a 2:2 fibre optic coupler which also optically mixes the LO reference signal with the received backscatter signal (as discussed in section 2.3.3). However, the 2:2 coupler has two outputs, thus giving two output signals which can be termed E_1 and E_2 [16]:

$$E_1 = \frac{1}{\sqrt{2}}(E_{RS} + E_{LO}), E_2 = \frac{1}{\sqrt{2}}(E_{RS} - E_{LO}) \tag{A.1}$$

Where E_{RS} and E_{LO} are the complex electromagnetic fields of the returned backscatter signal and the local oscillator respectively.

Thereafter, the two signals enter the balanced detector, which consists of two matched photodiodes and a transimpedance amplifier to convert their currents into a measurable voltage. The photodiodes respond with currents I_1 and I_2 respectively, proportional to the intensity of the electromagnetic field incident on it ($|E_1|^2$ and $|E_2|^2$ respectively), as was also discussed in section 2.3.3. Due to the internal setup of the transimpedance amplifier and the photodiodes in the balanced detector, the output current is then given by [16]:

$$I_{det} \propto I_1 - I_2 \propto |E_1|^2 - |E_2|^2 \propto 2\sqrt{P_{LO}P_{RS}}cos(2\pi\Delta ft) \tag{A.2}$$

Comparing this to Equation 2.5, it can be seen that the DC terms P_{LO} and P_{RS} are no longer present due to them being common-mode in both I_1 and I_2 and thus being canceled by the subtraction. This therefore lowers the noise floor, as the RIN contained in these terms is also eliminated.

Figure A.1 depicts this process graphically:

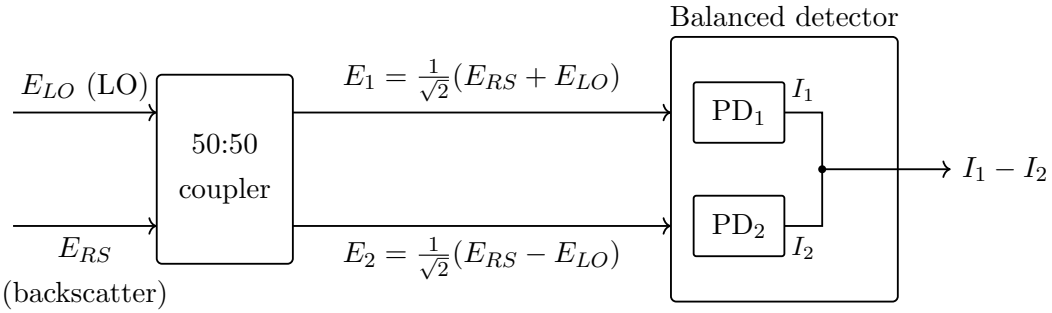


Figure A.1: Balanced detection. The 50:50 coupler delivers the sum and difference of the backscatter and LO fields to two matched photodiodes; the output is the difference of their photocurrents, in which the common-mode terms cancel and only the beat term $2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta ft)$ remains.

It can thus be seen that balanced detection provides the advantage of cancelling RIN. This allows for the final detected signal to theoretically be able to achieve the desired shot-noise limited operation (if other sources of noise are also eliminated).

A.2 A3: Supplementary Experiment on RIN Cancellation

Figure A.2 gives the 5 - 20MHz averaged PSD of the detected noise, when only a single photodetector input of the PD100B is used - effectively turning it into a singe-ended detector. The electronic noise floor will remain the same, but the total noise PSD averages will now also contain RIN in addition to shot-noise and electronic noise. Before analysis of this plot can commence, it is important to note that in the case of balanced detection, the shot-noise of each detector is uncorrelated, and will therefore add together at the output. Thus, in the single-ended case, the shot-noise at the output for a given laser current is expected to be half that of the balanced detection case. Any excess noise above this half-power shot-noise level is thus contributed from the electronic noise, and the uncancelled RIN.

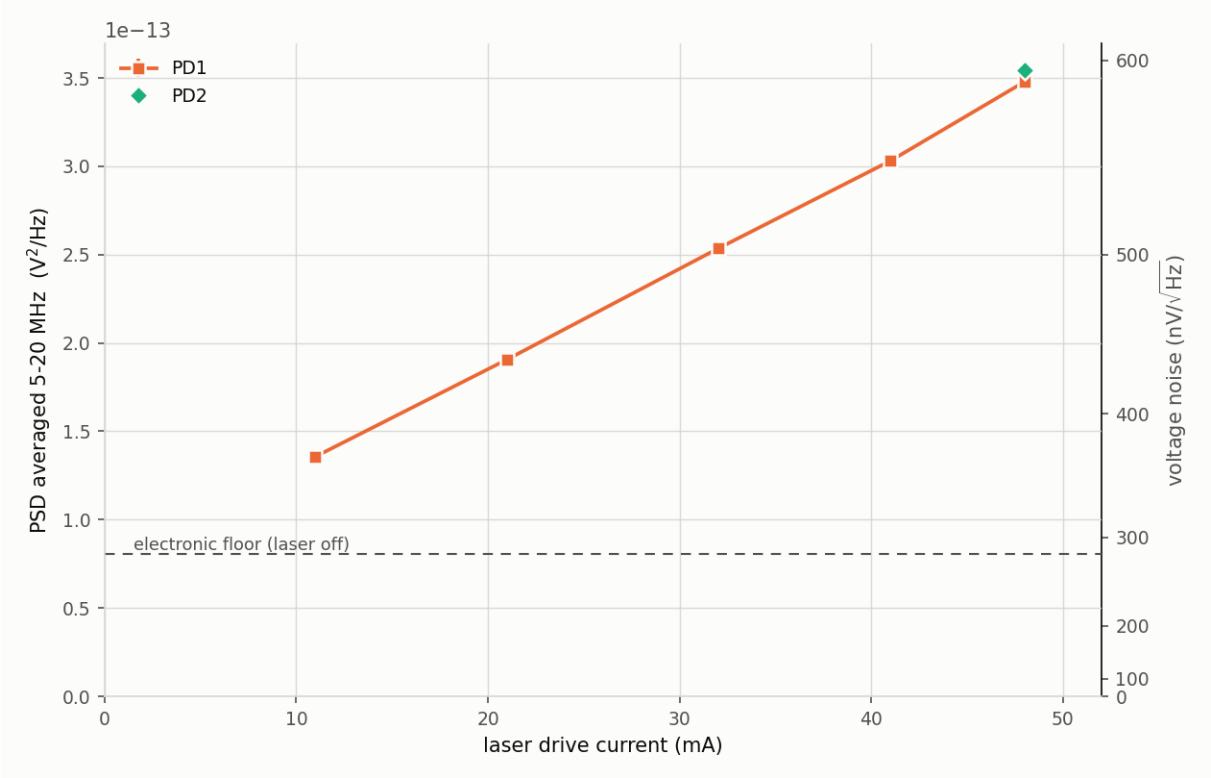


Figure A.2: A3: 5 - 20MHz Average PSD vs Laser Current with Single-Ended Detection

It can be seen in Figure A.2 that indeed, the total noise is lower than in Figure 4.3, as expected. Observing the total noise at the operating point of 48mA however, it can be seen that a value of approximately $3.5 \times 10^{-13} V^2/Hz$ is obtained, which gives $2.694 \times 10^{-13} V^2/Hz$ once electronic noise has been subtracted. In the balanced detection case at the same laser current, a value of $5.49 \times 10^{-13} V^2/Hz$ was obtained - subtracting electronic noise and halving this gives $2.342 \times 10^{-13} V^2/Hz$. This clearly indicates that the noise

power in the single-ended case is higher than simply half of the shot-noise in the balanced detection case, thus proving that indeed there is excess RIN present. This excess RIN noise can be calculated:

$$RIN_{excess} = (2.694 - 2.342) \times 10^{-13} = 0.352 \times 10^{-13} V^2/Hz \tag{A.3}$$

Converting to dB:

$$10log_{10}(1 + \frac{RIN_{excess}}{S}) = 10log_{10}(1 + \frac{0.352 \times 10^{-13}}{2.324 \times 10^{-13}}) = 0.6125dB \tag{A.4}$$

Thus, RIN contributes an additional 0.6125dB of noise above the ideal shot-noise limit at the 48mA operating point. Balanced detection allows this excess noise to be removed, as illustrated in the results of A2 - allowing for a better SNR.

A.3 Linewidth, PIIN and Doppler peak broadening

In order to understand the linewidth property of a laser source, it is important to first have an understanding of the concept of phase noise in a laser:

An ideal laser emits an electromagnetic wave $E(t) = E_0 cos(2\pi f_0 t + \phi)$ with ϕ a constant. A real laser emits $E(t) = E_0 cos(2\pi f_0 t + \phi(t))$ where $\phi(t)$ drifts unpredictably. This drift is called phase noise. Thus, when considering the frequency spectrum of the emitted laser beam over a long time period, it takes a Lorentzian shape centred around the nominal frequency of the laser (whereas an ideal laser would present simply as a delta spike at the nominal frequency). The laser linewidth, Δf, is then defined as the -3dB bandwidth of this Lorentzian spectrum, as illustrated in Figure A.3 below:

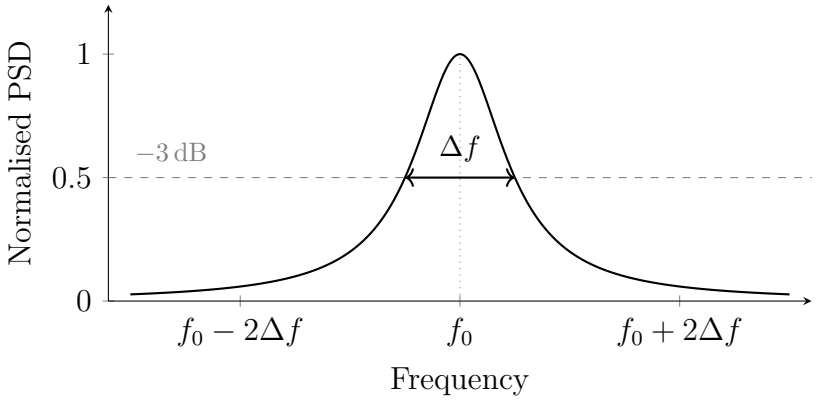


Figure A.3: Lorentzian frequency spectrum of a laser diode.

Asssociated with the linewidth is a quantity termed coherence length. The coherence length of the emitted beam, L_c, is given by [4]:

$$L_c = \frac{c}{2\pi\Delta f} \tag{A.5}$$

Where Δf is the linewidth of the source laser, and c is the speed of light. The coherence length is of relevance when considering linewidth effects.

The non-ideal linewidth property of a laser source gives rise to two primary effects in a DWL system's measurements - namely phase-induced intensity noise (PIIN), and the broadened Doppler peak:

As noted by Cariou et al. [7], Jiang et al. [15], and Harris et al. [11], reflections of the outgoing beam - caused by optics and the fibre optic transmit path - travel back in the system and are interfered with the reference beam (LO) before detection (in the same way as the actual backscattered signal of interest is). Because the reflection and LO derive from the same source, the mixing product is a strong, unwanted tone at baseband (homodyne) or at the intermediate frequency (heterodyne), carrying the phase noise accumulated over the path-length mismatch ΔL between the reflection and the LO. Harris et al. [11] show that the form of this residual noise is set by how large ΔL is relative to the coherence length (L_c) of the laser source.

For $\Delta L \gg L_c$ the two phases are highly uncorrelated because the difference in path (and therefore time) taken has allowed the phase of the two to "wander" significantly - and thus the tone broadens into a Lorentzian of width $2\Delta f$. This case is illustrated in Figure A.4 below, where it can be seen that this unwanted Lorentzian peak can serve to swamp out the desired Doppler peak to be measured - this is especially impactful for low wind-speeds, where the Doppler peak will be closer to the unwanted common-mode Lorentzian peak, and will be hidden more by it.

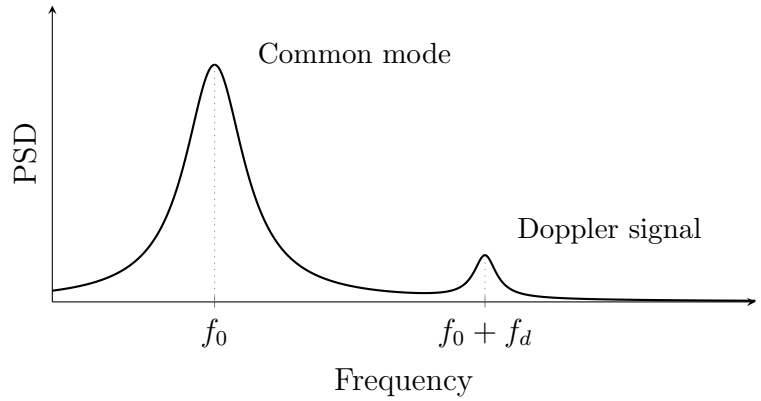


Figure A.4: Spectrum of a monostatic CW coherent lidar, showing the Doppler signal superimposed on the skirt of the common-mode peak (adapted from [7]).

For $\Delta L \ll L_c$, the case where the mismatch between LO and reflection paths is smaller, the reflected signal and the LO will be more correlated in phase. Thus, the residual instead appears as a flat excess noise floor across the frequency band, of level proportional to $\Delta f\, \Delta L^2$ [11]. This excess noise is termed excess LO noise, interferometric noise or phase-induced intensity noise (PIIN), depending on author [11, 1, 15]. For the remainder of this paper, this excess noise will be termed PIIN. PIIN has been measured at 5 - 30dB above shot noise in a diode-laser lidar [11] and well above shot noise in a fibre-laser system [15]. Thus, PIIN is a significant noise source which can cause the system to operate far from the ideal shot-noise limit discussed in subsection 2.3.5. It is thus clear that to reduce PIIN, a laser with as small a linewidth as possible should be selected, and the path length mismatch, ΔL L between LO and reflection paths should be minimised.

Another consideration arising due to non-ideal laser linewidth is the broadening of the measurable Doppler peak. This is particularly notable for longer ranges.

If the coherence length from Equation A.5 is long relative to the emitted signal's path length, the broadening of the Doppler peak due to the laser's linewidth may be neglected, and it will instead be other factors (such as turbulence in the air, and speckle) which dominate and cause a broader measurable peak [4]. This is because the total round-trip time of the beam is shorter, thus ensuring that the phase of the beam has not had "time" to shift significantly in between transmission, and later reception and mixing with the local oscillator. Thus, the received backscatter signal and the LO phase will be more coherent and will result in a sharper peak after mixing.

Conversely, if the coherence length is short relative to the signal path length (i.e. for targets at greater ranges), the Doppler peak will be spread over $2 \times \Delta f$, meaning that the measurement resolution and the ability to resolve turbulence in the air will be reduced [4].

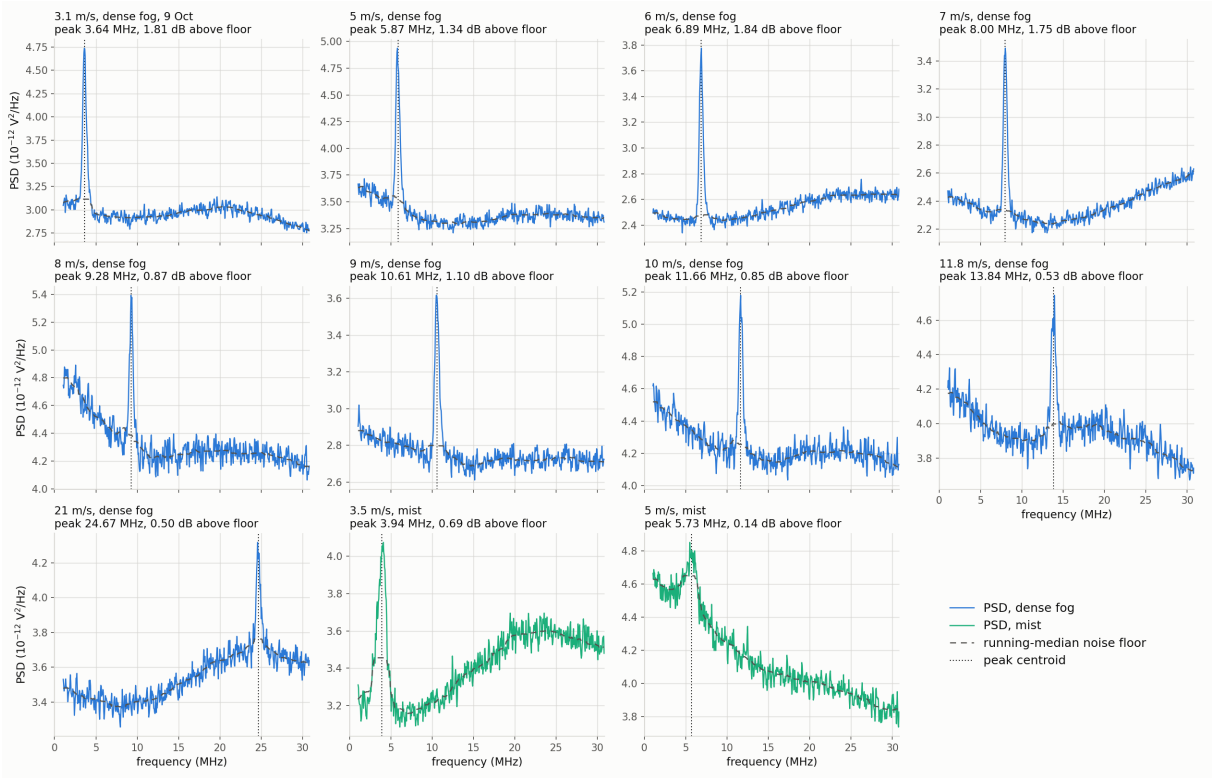


Figure A.5: Grid of PSD plots for each wind-tunnel setpoint

From the two effects described above, it can clearly be seen that a narrow linewidth laser source is highly beneficial for DWL systems, in order to achieve the best-case shot-noise limited operation, and to achieve a narrow, measurable Doppler peak.

IMPORTANT: Appendix A (see the table below) should provide a summary of how you have met the GAs associated with the course, and where in the report the evidence can be found.

If you have used AI in writing your report, you need to provide details in one of the appendices of how you used it.

If appropriate, in a subsequent appendix, provide a link to any GitHub repository where the code and additional materials for your project can be found.

A.4 D: Wind-tunnel tests full results

GA	**Requirement**	**Justification and section in the report**
1	Problem-solving	Description here
4	Investigations, experiments and data analysis	Description here
5	Use of engineering tools	Description here
6	Professional and technical communication (Long report)	Description here
8	Individual work	Description here
9	Independent learning ability	Description here

Appendix B

Addenda

B.1 Ethics Forms